CHAPTER 7

The Second Biological Psychiatry

In the 1970s, biological psychiatry came roaring back on stage, displacing psychoanalysis as the dominant paradigm and returning psychiatry to the fold of the other medical specialties. This triumph of the biological, the view that major psychiatric illness rested on a substrate of disordered brain chemistry and development, meant a return to themes that had last resounded in the nineteenth century at the time of the first biological psychiatry. It also entailed a repudiation of the psychoanalytic paradigm that saw psychiatric illness as psychogenic, arising in the mind as a result of faulty child rearing or environmental stress, curable through in-depth psychotherapy. There was room in the biological paradigm for psychotherapy, but it was the kind of informal psychotherapy inherent in the doctor-patient relationship and not the elaborate choreography of working through unconscious conflicts that psychoanalysis scripted.

Although the second biological psychiatry burst into the workaday world of clinical practice only in the 1970s, its views about genetics and brain development as causes, and its precepts about drugs and informal psychotherapy as remedies, germinated in the fading days of the first biological psychiatry. The immediate intellectual ancestor of the second biological psychiatry was the German Research Institute for Psychiatry in Munich that Emil Kraepelin, who wrote finis to the first biological psychiatry, had founded in 1917. Although Kraepelin himself was agnostic about causation—raising an eyebrow at the genetic and brain-biological views of his predecessors—he nonetheless surrounded himself with researchers such as Franz Nissl and Aloys Alzheimer who stressed the brain as the seat of psychiatric illness. Among these fellow researchers were several geneticists, who established at Kraepelin’s institute a laboratory for the study of inheritance patterns in psychiatric illness. It was early in the history of this lab, itself so ill-fated under the Third Reich, that genetic

patterns started to be traced establishing that the major psychiatric illnesses lay in the very substance of the brain itself and not in faulty patterns of mothering or an unhappy social environment.

The Genetic Strand

Among the most convincing kinds of evidence for a neural origin of major psychiatric illness would be genetic studies. A considerable portion of illness is noninheritable, the result of a developmental anomaly in the uterus or of environmental influences. Yet there is an inheritable, genetic portion as well, supporting the argument of biological psychiatry that nature plays at least something of a role.

The first biological psychiatry was mindful of this role, although its advocates, statistical naïfs, chose anecdotes rather than long series of patients balanced against control groups to support their case. It was also unhelpful to the historic reputation of the early biological psychiatrists that they cloaked their findings in the value-laden language of "degeneration." The precursors of the second biological psychiatry realized that to be convincing they would need quantitative data, comparing schizophrenic and depressed patients with controls. They also knew they must assemble data that would rule out the effect of family environment: It is always possible that children who become schizophrenic have somehow learned the behavior from sick relatives in the family environment rather than inheriting it.

That said, it is true that some of the first biological psychiatrists appreciated the importance of numbers. When Richard von Krafft-Ebing was still at Graz, he did a study of 19 psychotic female patients, of whom he found 12 to have some "neuropsychopathic family history."[1] As Emil Kraepelin reflected in 1913 on the charts of his Heidelberg patients, he noted that around 70 percent of those with schizophrenia had family histories of major psychiatric illness.[2] Thomas Clouston, master of the Edinburgh asylum, went about his task even more systematically. Whereas only 23 percent of all his patients had a family history of mental illness, 65 percent of those with "adolescent insanity" (his term for schizophrenia) did so.[3] Clouston was certain that these statistics embraced a genetic mechanism: "The ... view of the individual as being organically one in structure and

function with his ancestry and his posterity must always be kept in mind. They are not even links in a chain; links are separate from each other, and may have been forged from different pieces of iron; they are only one in function, whereas a man is just as much a part of his ancestry, and his posterity of him, as the root and stem are parts of one tree."[4]

The statistics of the nineteenth century all tended to confound genetic inheritance with family environment. To separate genes from environment, two approaches would be of service: twin studies and adoption studies. Not until the 1920s did the numerative skills of the international medical community achieve this level of sophistication.

The logic of a twin study is that identical, or monozygotic, twins develop from a single fertilized ovum and have a common set of genes. In the words of one British researcher, "This identical genetical inheritance must necessarily make them more alike, in all ways in which genes play a part, than are binovular [dizygotic] twins, who need resemble one another genetically no more than do ordinary brothers and sisters. Where differences do occur between the two members of an 'identical' twin pair it is the environment which is responsible."[5] But if there is a high degree of concordance, meaning a tendency for both of the identical twins to fall ill, it is likely that genetic influences are involved. This presumption of inheritance is even stronger if the concordance between the dizygotic twins is low. Studies from our own time show, for example, that the concordance between monozygotic twins in schizophrenia is about 50 percent, that for dizygotic twins about 15 percent.[6] This is one of the important pieces of evidence for the biological nature of schizophrenia. Francis Galton, who first proposed twin studies in 1875, called them an opportunity to "weigh in just scales the effects of nature and nurture."[7]

In 1928, Hans Luxenburger, a young Bavarian psychiatrist at the German Research Institute for Psychiatry in Munich, undertook the first large-scale survey of all twins born in a given area. He was trying to produce an unbiased series of twins rather than just concentrating on twins that happened to strike observers as curiosities. To establish the number of twin pairs in which one or both of the twins had a psychiatric illness, Luxenburger and colleagues asked all of Bavaria's mental hospitals to give them a list of patients present on a certain date. They then sent the list to all the local priests and pastors, asking if any of the listed individuals had been part of a twin birth. Among this series of 16,000 patients, the pastors were able to identify

211 as one of a twin pair. Going back to the patients' charts and doing further interviews of the hospitalized twin, the researchers made the diagnosis of schizophrenia in 106 cases. How often did the second twin have schizophrenia? The authors sought out the 65 second twins to have survived to adult life, interviewed them, and after much hemming and hawing about which twins were identical and which fraternal, found that in 7.6 percent of the monozygotic twin pairs both twins were ill, in none of the dizygotic.[8] This represented the first solid evidence that a major psychiatric illness—one considered "functional" by some (meaning cause unknown), psychogenic by others—had an organic substrate. Luxenburger, a staunch Catholic, was tarnished by association with his teacher and boss Ernst Rüdin, who later became a Nazi geneticist. Luxenburger himself served as an apologist for racist genetics for a few years under the Third Reich, then in 1941, as his criticisms of the regime endangered him, at Rüdin's suggestion he enlisted in the German Army, a place of relative safety from the Gestapo.[9]

Yet there was nothing intrinsically racist about the technique of twin studies in psychiatric genetics. Showing schizophrenia to be in part a genetic disease was really the high road of Central European scholarship. Manfred Bleuler, for example, who became professor of psychiatry in Zurich in 1942, was a mainstream scholar by any stretch. Yet when a visiting American researcher came calling at the door of the Bleuler home, Manfred met him waving "an old display case of butterflies" showing crosses and backcrosses in the wings. It has been Bleuler's high school science project. Bleuler told the visitor "that it had been his lifelong dream to discover the Mendelian basis of schizophrenia."[10]

So the undertaking was not Nazi-inspired. Indeed, the next major contributions to the field came from Jewish scholars. Aaron Rosanoff, born in Russia in 1879, came to the United States at the age of 13, graduated in medicine from Cornell University in 1901, then entered the New York State asylum system at King's Park State Hospital on Long Island. In a system known for attracting benchwarmers and time-servers, Rosanoff distinguished himself for his scientific curiosity, especially about medical genetics. After moving to Los Angeles in 1922 to open a private nervous clinic (the Alhambra Sanitarium), he resolved to undertake a large-scale twin study using data from state hospitals.[11] By the early 1930s, he had his numbers in place. Because the small number of twin pairs had been Luxenburger's Achilles' heel, Rosanoff compiled a list of 1,014 twin pairs in which one of

the twins had a major mental disorder. Of these 1,014 pairs, schizophrenia was diagnosed in 142. Among the monozygotic twins, both twins developed schizophrenia in 68.3 percent of the cases, among the dizygotic, in 14.9 percent.[12] The difference was enormous. Rosanoff also looked at manic-depressive illness, where in 90 twin pairs at least one twin had the illness. The second twin got it as well in 69.6 percent of the monozygotic twin pairs, in only 16.4 percent of the dizygotic. Rosanoff concluded, ”Hereditary or germinal factors play an important part in the etiology of manic-depressive syndromes.“[13]

It must be emphasized that Rosanoff was not a marginal right-wing figure, as some historians of psychiatry have tried to characterize these early psychiatric geneticists.[14] After becoming director of the California mental-hospital system in 1939, he inaugurated a statewide program for the home-care of patients, and was instrumental in establishing a Maudsley-style research center in San Francisco, the Langley Porter Clinic. Rosanoff’s twin studies arguably represent the major American contribution to international psychiatric literature in the years between the two world wars, yet the official histories of American psychiatry, dominated by psychoanalytically oriented writers, pass over his work in virtual silence.[15]

One of the coworkers in Ernst Rüdin’s and Hans Luxenburger’s genetics lab was the young Franz Kallmann. Born in Neumarkt, Silesia, in 1897, Kallmann earned his MD in 1919 at Breslau University, then trained in psychiatry at Kraepelin’s research institute in Munich. In 1929, Kallmann became involved in the large-scale studies that Rüdin and Luxenburger were undertaking, and as an extension of that work moved to the Herzberge Mental Hospital in Berlin where he did a family study of all blood relatives of schizophrenics admitted to the hospital 30 years previously. The Nazi racial laws of 1935 obliged Kallmann to give up his post in Berlin, whereupon he took his notes across the Atlantic to the New York State Psychiatric Institute. In 1938, he published in English the results of the Berlin study (which did not include enough twins to make an interesting analysis).[16]

In the meantime, however, Kallmann had seen Rosanoff’s work. In the early 1940s, Kallmann decided to apply his tremendous energy to a twin study of all patients in public asylums in New York State. Of the 73,000 patients in the system in 1945, Kallmann identified 691 schizophrenics with a traceable cotwin. For the monozygotic twins, the concordance rate was 85.8 percent, for the dizygotic 14.7.[17] Thus in identical twins there was an almost

certain chance of the second twin developing the illness, in fraternal only a one in seven chance. These findings were such a solid index of organicity in schizophrenia that, when they were presented in 1950 at the First World Congress of Psychiatry in Paris, they caused a sensation. (Eliot Slater read Kallmann's paper.) Kallmann had written in an unvarnished manner about the Mendelian inheritance of schizophrenia (meaning a single locus in the DNA), and such was the storm of protest from delegates assembled from around the world that, to "de-dramatize" the discussion, the chairman refused to read from any of the written comments that had been sent in. Most opposed were the psychoanalysts, whom Kallmann had dubbed "a host of cynical armchair workers."[18]

Other large-scale twin studies were done after World War II[19], yet it was by now apparent that schizophrenia and manic-depressive illness were heavily genetic in nature. The closer the blood relationship, the higher the rate of schizophrenia in other family members.[20] Yet could the family environment play a role as well? After all, monozygotic twins grew up together. Perhaps living in a family environment that was "distorted"—in the phrase of the day—by mental illness made other family members ill. How much did environment contribute as opposed to heredity? How many patients had been brought low by a schizophrenogenic mother, a flawed family dynamic?

In 1959, Seymour Kety decided to look at children of schizophrenic mothers or fathers who had been raised in foster homes, thus eliminating the effect of growing up midst madness. Kety himself was one of the founders of biological psychiatry in the United States. Born in Philadelphia in 1915, he received his MD degree from the University of Pennsylvania in 1940, going on to do research at that university and teach the pharmacology and physiology of the brain. In 1951, Kety became the scientific director of the recently founded (in 1949) National Institute of Mental Health in Bethesda, shifting the institute's course from research on psychoanalysis to research into basic science. He remained at NIMH until becoming a professor at Harvard in 1967. After retiring from Harvard in 1983, Kety returned to Bethesda to continue directing one of the meatiest research projects in the history of psychiatry: the Danish adoption study. The state of Denmark has an exceptional capacity to follow its citizens through life. An adoption register of the state Department of Justice permits researchers to identify the biological relatives of children who are adopted; a population register makes

it possible to locate adoptees and follow their trajectories through life. Thus in Denmark, one is able to study the biological background of adoptees as well as their fates in their new families. In 1968, Kety and collaborators published the results of their study of 5,483 adoptions in Copenhagen between 1924 and 1947. From this group, 507 adoptees were later admitted to a psychiatric hospital. Independent observers reviewed their case histories and identified 33 of them as schizophrenics. These patients and their families were then compared with an age-matched control group of adoptees never admitted to a psychiatric hospital. Within the biological families of the adopted schizophrenics, about 10 percent of the close relatives had schizophrenia, within the families of the control group there was very little schizophrenia. Nature and not just nurture had contributed to making these adopted children ill. Kety and colleagues concluded cautiously that "genetic factors are important in the transmission of schizophrenia." The mechanism must involve a number of genes rather than a single-gene Mendelian model as Kallmann had posited.[21]

By 1992, Kety had widened the study to include adoptees in all of Denmark, learning that almost half of the schizophrenic adoptees had a history of some form of the disease in their biological families, versus virtually none of the controls.[22] For all of Denmark, schizophrenia was 10 times more common in the biological relatives of the schizophrenic adoptees than in the biological relatives of the controls. It was also found more often in the brothers and sisters of the adoptees (12.5 percent) than in second-degree relatives (2.2 percent), which is "also consistent with genetic transmission," the authors noted.[23]

Kety's research on adoptees with schizophrenia unleashed an avalanche of genetic research involving family, twin, and adoption studies. In 1977, a Danish study of manic-depressive illness found concordances of 67 percent for monozygotic twins and 20 percent for dizygotic.[24] Twin studies in the 1980s implicated genetic factors in agoraphobia and panic disorder.[25] An adoption study of psychosomatic illness highlighted genetic factors: In families where the children of violent and alcoholic fathers were placed in foster homes, the boys tended to behave like their biological fathers (even though the father had no contact with them); the daughters of such biological fathers tended to display abnormal behavior also, the kind of chronic physical complaints of nonorganic origin that were once known as "hysteria."[26] Other kinds of research suggested that such hysteria is passed down in the female

line at the same time as antisocial behavior in the male line.[27]

Twin studies soon began bearing fruit across the entire terrain of psychiatry. In such personality traits as hypochondriasis, hypomania, and depression, psychological testing unearthed significant differences between monozygotic and dizygotic twins—the litmus test of the genetic strand.[28] This was poaching on the classic terrain of psychogenesis: For a century, the doctrine had ruled that hysteria and the symptomatic psychoneuroses were the result of stress or dysfunctional family life. The genetic news implied that these illnesses must have a significant brain substrate, however much environmental circumstances might contribute to triggering them.

Yet what of the family? If overall, genetics accounted for about 50 percent of such "behavioral disorders" as schizophrenia, was the family responsible for the other half? Not at all. Studies of pairs of adopted children, themselves unrelated but raised in the same foster family, showed correlations close to zero. As one study put it, "the relevant environmental influences are not shared by children in the same family."[29] Whatever had made the one adopted child sick had no effect on the other. Thus the environmental influence could not have been family life itself.

In the 1970s and after, genetic psychiatry became even bolder in the search for causes, as molecular biology opened the possibility of identifying the actual genes involved. By 1995, the gene or genes causing schizophrenia had been tentatively placed somewhere on chromosome 6.[30] For manic-depressive illness, chromosomes number 18 and 21 had been implicated. [31] Psychiatric geneticists began to propose genetic anticipation, the tendency for some illness-causing genes to expand in size when passed from generation to generation, as the mechanism behind the increasing severity of schizophrenia or manic-depressive illness as handed down a family tree.[32] This was the exact equivalent of "degeneration" in the first biological psychiatry. The schizophrenogenic mother was truly dead.

The First Drug That Worked

The other wing that carried psychiatry into the biological era was drug therapy. Drugs had always been used in psychiatry, from the laxatives that

were once administered to patients newly admitted to asylum—on the grounds that their problems might be caused by colonic autointoxication—to opium and its alkaloids for depression and mania: mildly successful but highly addictive. The modern era of drug therapy began with systematic experimentation into the chemistry of the brain. Brain chemistry means neurotransmitters, the chemicals that transmit the nerve impulse from one neuron to another across the synapse (the gap between the neurons). Although research on brain chemistry went back to turn-of-the-century English physiologists, it was only in the early 1920s that Otto Loewi, professor of pharmacology at the University of Graz, isolated the first neurotransmitter. On the basis of work he began in the winter of 1921, in 1926 he was able to say that the chemical acetylcholine mediated the transmission of the nerve impulse from one nerve to the next.[33]

The discovery of acetylcholine did not remain abstract knowledge but had rapid therapeutic consequences: In the 1930s, psychiatrists gave acetylcholine to their patients in the hopes of relieving schizophrenia, although at this point they had no notion of the mechanism involved.[34]

Psychosis before the introduction of chlorpromazine: Patient at a German mental hospital around 1900 listening to his wife's voice from an alarm clock (courtesy of the Universitatsbibliothek Leipzig).

Buoyed by the success of the physical therapies of the 1930s, many mental-hospital psychiatrists were open-minded toward the potential of drugs, and throughout the 1940s experimentation went on relentlessly in asylums to find agents that would improve on shock and coma therapies:

> In 1937, Heinz Lehmann, a refugee psychiatrist from Berlin, had just arrived at Montreal's Verdun Protestant Hospital, where he was one of a few physicians for some 1,600 psychiatric patients. "It was pretty horrible to work under those conditions," Lehmann said. "So I did all kinds of things, always convinced that psychotic conditions and the major affective disorders ... had some sort of a biological substrate. I kept experimenting with all kinds of drugs, for instance, large doses, very large doses of caffeine, I remember, in one or two stuporous catatonic schizophrenics—of

> course, with no results." He injected sulfur suspended in oil into his patients, "which was painful and caused a fever." He injected typhoid antitoxin to produce a fever analogous to the malaria therapy. "Nothing helped; I even injected turpentine into the abdominal muscles which produced—and was supposed to produce—a huge sterile abscess and marked leucocytosis [raising the white count]. Of course, that abscess had to be opened in the operating room under sterile conditions. None of this had any effect, but all of this had been proposed in, mostly, European work as being of help in schizophrenia."[35]

The point is not that researchers such as Lehmann behaved inhumanely with their patients: They were searching in the best of faith for something better to offer them. It is rather that, by the time in 1951 that Henri Laborit, a surgeon in the French navy, began experimenting with a curious new "potentiator" of anesthetics, the ground had already been well prepared for the reception of new antipsychotic drugs.[36] Laborit himself was directly responsible for a drug that changed the face of psychiatry: chlorpromazine.

Stationed at the Maritime Hospital at Bizerte in Tunisia in 1949, Laborit was 35 when he began researching various synthetic antihistamines as a means of "potentiating" anesthetics. Operating on soldiers in shock has always been a big problem in military surgery, and Laborit's idea was that a potentiator could block the autonomic mechanisms involved in shock, increasing the chances of success of the operation. Among possible potentiators, Laborit was working with some recent antihistamines in the phenothiazine family synthesized by the Rhône- Poulenc drug company. Ever since the discovery of antihistamines in 1937, physicians had been experimenting with these drugs on psychotic patients, with little result. At this point, Laborit had no interest in psychosis, but he did note that some of his surgical patients became quite indifferent to the world about them (ataraxic was the term subsequently applied) after being given the phenothiazines. As he later said, "I asked an army psychiatrist to watch me operate on some of my tense, anxious Mediterranean-type patients. After surgery, he agreed with me that the patients were remarkably calm and relaxed. But I guess he didn't think any more about his observations, as they might apply to psychiatric patients."[37]

Early in 1951, Laborit was transferred from the marine hospital system, where interest in his work was virtually nonexistent, to the physiology

laboratory of the Val-de-Grâce military hospital in Paris. Here, surrounded by inquiring minds and under the guidance of a benevolent lab chief, he pressed ahead with his work on shock. Because the previous antihistamines had not been ideal autonomic blockers, in June 1951 Laborit asked Rhône-Poulenc for a sample of a new phenothiazine that company chemist Paul Charpentier, a phenothiazine specialist, had recently synthesized, dubbed 4560 RP (for Rhône-Poulenc). Charpentier later called the compound "chlorpromazine." Even before Laborit's request, the company suspected that it might be a useful psychiatry drug but had not yet sent it to clinicians for trials.[38]

Laborit began using 4560 RP on surgical patients at Val-de-Grace. He found that, in addition to doing what it was supposed to surgically, the drug produced a certain "uninterest" *(désintéressement)* on the part of the patient. In the tradition of antihistamine researchers who had tried to apply their compounds psychiatrically, Laborit decided to do so as well, and in November 1951 he tested it for toxicity on a colleague, a female psychiatrist, at the Villejuif Mental Hospital. Shortly after taking it intravenously she got up to go to the bathroom and fainted. The chief of the psychiatry service at Villejuif thereupon resolved to have nothing more to do with it.[39] So Laborit returned to Val-de-Grâce and, over lunch at the hospital canteen, finally persuaded three rather unenthusiastic psychiatrist colleagues to give it to their patients.[40] In early February 1952, Laborit reported to the medical press the results of his surgical work with 4560 RP, dropping out of the blue the suggestive line at the end of the article, "These findings allow one to anticipate certain indications for the use of this compound in psychiatry, possibly in connection with barbiturates in a deep-sleep cure."[41] Yet as he penned those lines, perhaps tongue in cheek, on January 19, 1952, the Val-de-Grâce psychiatrists were giving chlorpromazine to Jacques L., a 24-year-old patient with an attack of mania. "The patient is calm after the injection. He lies without moving, his eyes shut, responds when spoken to, with a maniacal look on his face; he closes an eye, pulls on his tongue, then sleep." He woke up manic again. Over the next three weeks he received chlorpromazine in combination with an analgesic, a barbiturate, and ECT. Finally by February 7, he was calm enough that he could play bridge and lead a normal life, though his attitudes remained somewhat "hypomanic."[42] What had moderated the patient's symptoms was not at all clear, but at least chlorpromazine hadn't harmed him, and in the meantime news was racing about the Paris grapevine that a big new psychiatry drug was in the works.

Word of chlorpromazine reached the ears of two of the big guns in Parisian psychiatry, Jean Delay and Pierre Deniker at the Ste-Anne mental hospital, who began giving it to their patients in March 1952.[43] Delay, 45 at the time, was professor of psychiatry at the Sorbonne and director of the Ste-Anne. Deniker, 10 years younger, was on the hospital's medical staff. At the prestigious hundredth anniversary of the Medical Psychological Society in May 1952, they described briefly their work on chlorpromazine, omitting any reference to Laborit.[44] Then at the Society's June meeting, they delivered a fuller report on eight patients.[45] While it is true that they were the first to treat a series of patients exclusively with chlorpromazine, it is not true that they discovered the drug's psychiatric applications. And history has wrongly exalted Delay and Deniker in connection with opening this epochal new chapter in the annals of psychiatry, the advent of pharmacotherapy, while sliding in silence over that of Laborit.[46]

Nonetheless, by May 1952 it was evident that Delay's and Deniker's patients at the Ste-Anne had done well on chlorpromazine. Patient number one, Giovanni A., a 57-year-old laborer with a long history of mental illness, had been admitted most recently for "making improvised political speeches in cafés, becoming involved in fights with strangers, and for the last few days has been walking around the street with a pot of flowers on his head preaching his love of liberty." After nine days on chlorpromazine, he was able to have a normal conversation, after three weeks he had calmed and was about to be discharged. The results for the seven other patients were similar.[47] This was much better than ECT, insulin, and the rest of the physical therapies, much less dangerous, and easily tolerated by the patients.

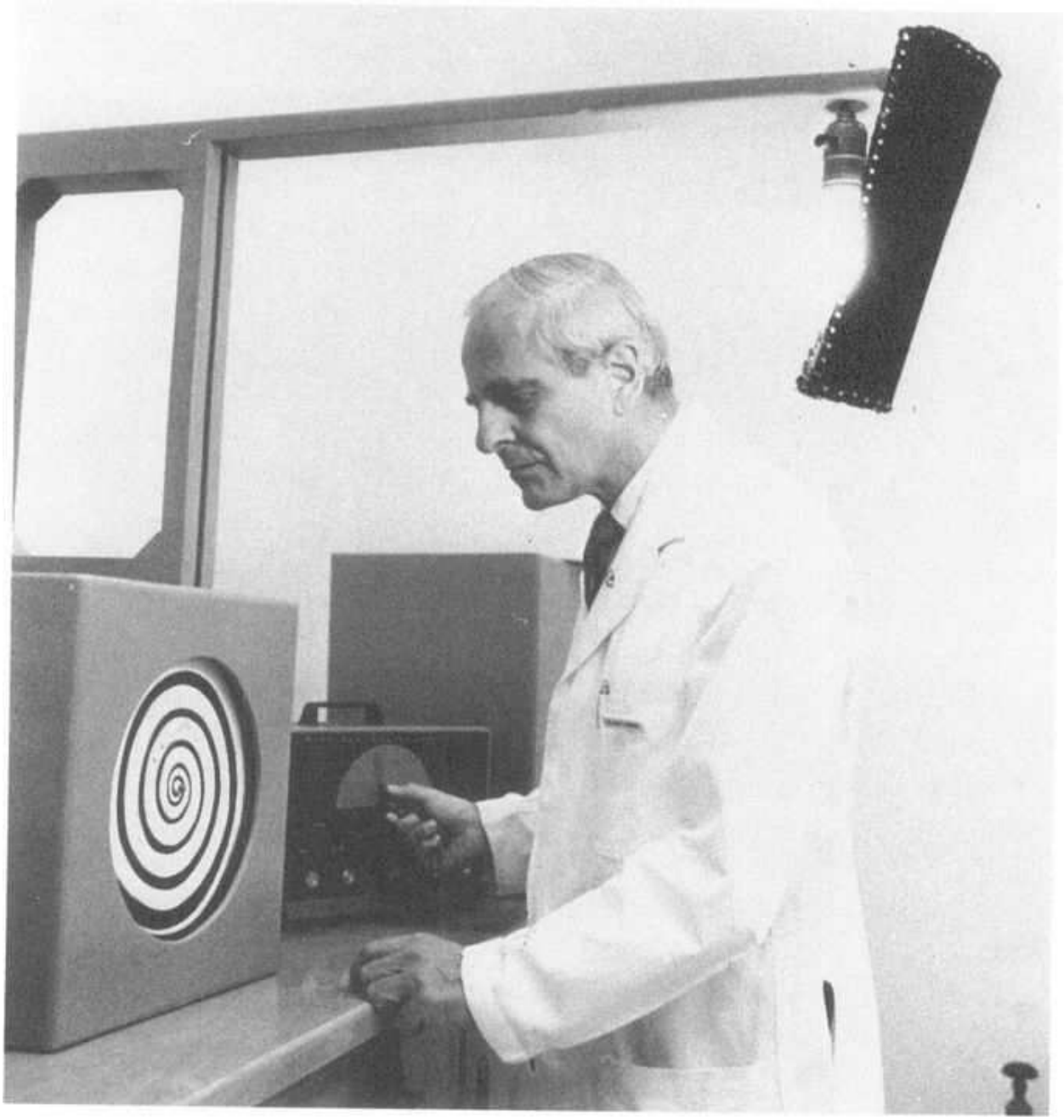
Chlorpromazine spread at once through the French system. "By May 1953," writes one historian, "the atmosphere in the disturbed wards of mental hospitals in Paris was transformed: straitjackets, psychohydraulic packs and noise were things of the past! Once more, Paris psychiatrists who long ago unchained the chained, became pioneers in liberating their patients, this time from inner torments, and with a drug: chlorpromazine. It accomplished the pharmacologic revolution of psychiatry."[48]

The ultimate force behind the discovery and adoption of new drugs such as chlorpromazine was not academic scientists or clinicians such as Laborit and Delay but the drug companies. Although Rhône- Poulenc had not identified chlorpromazine as an antipsychotic, it was company scientists who had

systematically designed the compound and tested it on animals.[49] The drug's discovery owed nothing to serendipity. As chlorpromazine made its way into the world, the driving force would be pharmaceutical executives and scientists, a point worth recalling because, in the decades that followed chlorpromazine, the race to discover new drugs would be dominated by the great pharmaceutical houses.

Clinicians brought chlorpromazine to North America but a drug house gained its acceptance. In 1952, young Dr. Ruth Koeppe-Kajander, who had earned her MD at Göttingen just after the war, was interning at the Oshawa (Ontario) General Hospital when she noted a staff anesthetist giving chlorpromazine (or Largactil, its trade name outside the United States) to potentiate the anesthesia. The anesthesist had received it on trial from Rhône-Poulenc. When Koeppe's internship ended in 1953, she began training in psychiatry at the mental hospital in London, Ontario, and—as a first-year resident—gained permission to give the drug to 25 patients over a period of months. In November 1953, she reported "on the remarkable effects of this drug" at a psychiatric meeting near Toronto. "It calmed restless, excited or overactive patients without sedating them to the level where they could not function. Patients lost their agitation but not their consciousness. They could talk about themselves and eat and sleep without difficulty. Catatonic and other types of excitement were no longer life threatening."[50]

Heinz Lehmann of the Verdun Protestant Hospital in Montreal, who introduced chlorpromazine to North America in 1953. Here, in a photograph from the 1960s, Lehmann is using an Archimedes spiral to measure the effect of psychotropic drugs (courtesy McGill University Archives).



Yet the name associated with the introduction of chlorpromazine in North America is not Koeppe-Kajander but Heinz Lehmann at the Verdun Hospital in Montreal. French Canada was a logical beachhead because Rhône-Poulenc had an office in Montreal. “One day,” recalled Lehmann, “there was a detail man [sales representative] from ... Rhône-Poulenc, and he left all kinds of literature and samples. My secretary told him that I was much too busy to see him, but he said, ‘It isn’t necessary, I’ll leave this here, this is something new

and so good I don't have to explain it to him, he will certainly pay attention to it once he reads it.'"

Though Lehmann thought this rather arrogant, it did catch his attention. That Sunday as he lay in the bathtub he read several of Deniker's papers. Lehmann, a German, could read French easily because he had a French-Canadian wife and spoke French at home (the other hospital psychiatrists were all unilingual in English). At first, Lehmann thought it was just another sedative. "But there was a certain statement: it acted 'like a chemical lobotomy,' which puzzled me, and I said to myself, there is something more to it. Obviously these two, Delay and Deniker, are very sophisticated psychiatrists, they must know what they are talking about." Lehmann asked for several nurses to act as volunteers and gave them small oral doses, which made them drowsy but didn't alter any other intellectual functions, unlike the barbiturates. Lehmann got a supply of chlorpromazine from the company sufficient for 71 patients, and lined up a resident to help him.

The results astonished him. As he later said, "Two or three of the acute schizophrenics became symptom-free. Now I had never seen that before. I thought it was a fluke—something that would never happen again but anyway there they were. At the end of four or five weeks, there were a lot of symptom-free patients. By this, I mean that a lot of hallucinations, delusions and thought disorder had disappeared. In 1953 there just wasn't anything that ever produced something like this—a remission from schizophrenia in weeks."[51]

Between May and July 1953, he gave the patients the drug and recorded the results, writing up his findings as a "unique" new therapeutic agent.[52] The director of the asylum told him, "You never use the word 'unique' in anything that you publish, because you always regret it later on—there is no such thing as unique." Lehmann left the "unique" in.[53]

All kinds of things began happening. Said Lehmann, "Chronic schizophrenics who had been divorced because they had been psychotic for 10 years, now all of a sudden they were symptom-free and their husbands or their wives were married again. It was a very strange time."[54]

There was one more strange thing. As Lehmann and a colleague stood one day observing the patients, they noted that some "walked with a peculiarly stiff gait (they had that peculiar mask-like face) and we said that looks like Parkinsonism, but it did not seem possible because at that time there was no such thing as drug-induced Parkinsonism.... There were two of these patients,

and then a couple of weeks later another two or three, and we then named these side effects extrapyramidal symptoms."[55] (The pyramidal tract in the spinal cord gives voluntary control of muscles; involuntary movements must be "extrapyramidal.") The symptoms would later be called tardive dyskinesia. What seemed in 1953 a rather innocuous side effect would turn out to be a massive social and medical problem, as patients from these mental hospitals were, in a few years time, discharged en masse to the streets: To avoid these and similar side effects they would stop taking their medication.

Yet in 1953, the issue of involuntary grimacing and embarrassing, uncontrollable bodily movements lay in the future. What counted was Lehmann's forceful presentation to an English-speaking audience of chlorpromazine as the first drug in psychiatry to abolish the symptoms of psychosis, though not necessarily to cure the underlying brain disorder. The images of rapturous release from illnesses such as mania were impossible to resist: "One manic patient stated after her recovery that with the drug she soon lost the feeling: 'I had to live my whole life in one day.'" Another patient suffering from anxiety said, "It was like a chairman taking control of a meeting where everybody had been shouting at once."[56] These arresting lines in a major medical journal guaranteed a continentwide hearing for the new drug.

The scene shifts to the United States, toughest market of all to crack because of the predominance of psychoanalysis and its predilection for getting to the "real" causes of the illness. Here an ambitious young drug house named Smith Kline & French enters the picture. The company was a maker of patent medicines, and its new president, Francis Boyer, wanted to upgrade it to a manufacturer of "ethicals," meaning drugs prescribed by the medical profession. Aware that Rhône-Poulenc had a hot new potentiator going—but unaware that it might have psychiatric uses—in the spring of 1952 Boyer went to France. Rhône-Poulenc was very close-mouthed. (The company had already tried unsuccessfully to interest several other big American firms.) When Boyer signed the licensing agreement, he thought he was buying an antiemetic (anti-vomiting drug). Having virtually no research budget, Smith Kline was not prepared to undertake extensive trials. Said Boyer, "Let's get this thing on the market as an antiemetic and we'll worry about the rest of that stuff later." The company brought it out as "Thorazine."

Many years later, this author interviewed John Young, who at the time was a 29-year-old member of Smith Kline's international division. When did

Boyer realize that chlorpromazine was a major antipsychotic?

"Dr. Shorter, you're not old enough to remember that's a meaningless question because antipsychotics didn't exist. There were no drugs. Those cases were in the back wards and that was it. The notion that you could ever do anything about it hadn't occurred to anyone. These doctors and nurses who went into McLean [Hospital] at that time, their altruism was busted after six months. What this drug did was not to cure these patients but to tip the balance enough so that the doctor and the nurse in the hospital said, 'Hey, maybe we can do something for these people.'"[57]

The first psychiatrist to try Smith Kline's chlorpromazine on patients was William Long, the firm's medical director. One of his first five patients, all with mania, was a "severely disturbed nun," as Young recalled, "at the edge of violence and using extremely coarse language. He was very concerned about the patient. He gave her some of this stuff. The results? He couldn't believe it. She had been extraordinarily abusive with most un-nun-like behavior. In the afternoon she was calm. He described it at the table in the lunchroom. What he described was a typical chlorpromazine result."

It was important for Smith Kline to get outside psychiatrists to run drug trials, not an easy task given the prevailing climate of opinion in American psychiatry. In September 1953, psychiatrist Willis Bower at McLean began the trial that would prove most influential. He reported the drug to be "capable of strongly influencing the course of some psychiatric illnesses." There was no memory loss, as with ECT, and unlike the barbiturates it did not disinhibit patients nor sedate them. The study was published in the New England Journal of *Medicine*.[58]

Having ensured chlorpromazine's acceptance in academic medicine, Smith Kline formed a special chlorpromazine task force and proceeded to beat the bushes in the state asylums, the elective terrain of the drug. If the task force encountered a reluctant state hospital, it would point out to state legislators the money-saving aspects of using an antipsychotic drug.[59] Thus chlorpromazine, though scorned by analysts such as Jules Masserman as a "glorified sedative," became widely adopted in the red-brick state hospitals. Time magazine later mocked the Freudians for their airs: "The ivory-tower critics argue that the red-brick pragmatists are not getting to the patient's 'underlying psychopathology' and so there can be no cure. These doctors want to know whether he withdrew from the world because of unconscious conflict over incestuous urges or stealing from his brother's piggy bank at the

age of five. In the world of red bricks, this is like arguing about the number of angels on the point of a pin."[60]

But it wasn't just in the snake pits that chlorpromazine found a use. It damped the delusions and hallucinations and calmed the agitation of patients in such expensive private-sector clinics as the Sheppard and Enoch Pratt Hospital in Baltimore. With chlorpromazine and other drugs, "the wild, screaming, unapproachable patients [became] a thing of the past," wrote one staffer. "Many more patients could go for drives in the country, visit Towson and Baltimore for shopping excursions, with or without attendants, go to the theatre, visit art museums, take in athletic contests, and go out with relatives for dinner. Life became more varied and interesting, and improvement was advanced."[61]

Chlorpromazine initiated a revolution in psychiatry, comparable to the introduction of penicillin in general medicine. While it did not cure the diseases causing psychosis, it did abolish their cardinal symptoms so that patients with underlying schizophrenia could lead relatively normal lives and not be confined to institutions. In 1955, Delay and Deniker proposed the term "neuroleptic" for drugs that diminished psychosis, though Americans preferred the term "antipsychotic."[62] English psychiatrist Henry Rollin, who had worked at the big asylum at Epsom, later said that chlorpromazine "tore through the civilized world like a whirlwind and engulfed the whole treatment spectrum of psychiatric disorders." It was the beginning of "the era of psychopharmacology."[63]

The Cornucopia

Following chlorpromazine, a veritable cornucopia of antipsychotic, antimanic, and antidepressant drugs poured forth, changing psychiatry from a branch of social work to a field that called for the most precise knowledge of pharmacology, the effect of drugs on the body. Of course, all that spilled from the cornucopia was not bounty. Some were me-too drugs, thrown onto the market for competitive reasons only; others were identified as toxic and soon withdrawn; still others lapped from psychiatry into the street as drugs of abuse. Yet on the whole the products of the cornucopia greatly ameliorated the lot of individuals with mental illness. It was with these psychiatric drugs

that the journalist's cliched phrase "new hope" originated.

The story began in 1949 with John Cade, the 37-year-old superintendent of the Repatriation Mental Hospital in Bundoora, Australia. Cade, like Neil Macleod in late-nineteenth-century Shanghai, had not lost his scientific curiosity despite his provincial isolation. He was determined to see if the cause of mania was some toxic product manufactured by the body itself, analogous to thyrotoxicosis from the thyroid. Not having any idea what, exactly, he might be searching for, he began taking urine from his manic patients and, in a disused hospital kitchen, injecting it into the bellies of guinea pigs. Sure enough, the guinea pigs died, as they did when injected with the urine of controls. Cade began investigating the various components of urine—urea, uric acid and so forth—and realized that to make uric acid soluble for purposes of injection he would have to mix it with lithium, an element that had been used medically since the middle of the nineteenth century (in the mistaken belief that it could serve as a solvent of uric acid in the treatment of gout).

Then Cade, on a whim, tried injecting the guinea pigs with lithium alone, just to see what would happen. The guinea pigs became very lethargic. "Those who have experimented with guinea pigs," he wrote, "know to what degree a ready startle reaction is part of their makeup. It was thus even more startling to the experimenter to find that after the injection of a solution of lithium carbonate they could be turned on their backs and that, instead of their usual frantic righting reflex behavior, they merely lay there and gazed placidly back at him."

Cade had stumbled into a discovery of staggering importance, yet he was able to develop it only because of his resoluteness in taking the next step. He decided to inject manic patients with lithium. First he injected himself to see if repeated doses of lithium citrate and lithium carbonate would be harmful to him, which they were not. (Cade probably had a rather high tolerance of discomfort given that he had spent three years in a Japanese prisoner-of-war camp.) Then he injected 10 of his manic patients, 6 schizophrenics, and 3 chronic psychotic depressives. The lithium produced no impact on the depressed patients; it calmed somewhat the restlessness of the schizophrenics. But its effect on the manic patients was flamboyant: All 10 of them improved, though several discontinued the medication and were still in hospital at the time Cade wrote his article late in 1949. Five were discharged well, though on maintenance doses of lithium.[64]

Cade was lucky that he gave the lithium to patients with low-grade chronic mania rather than to those with full-blast mania, who are unresponsive to the drug. Yet not only was the serendipitous method of the discovery stunning, the results themselves were positive almost beyond belief. Consider patient number eight, a man of 50 years who had suffered recurrent mania attacks since the age of 20. “The present attack had lasted two months and showed no signs of abating. He was garrulous, euphoric, restless and unkempt when he started taking lithium citrate 20 grains three times a day on February 11, 1949.” By the ninth day, he had commenced work in the garden. “By the end of two weeks, he was practically normal—quiet, tidy, rational, with insight into his previous condition. This was in marked contrast to his condition a fortnight before when he had to be locked in a single room at night with a regular nocturnal hypnotic and was too restless to eat in the dining room owing to his unsettling effect on the other patients.”

The year 1949 was the worst possible time to bring out a paper on the benefits of lithium. In that year, the *Journal of the American Medical Association* published cases of two patients in congestive heart failure who died after being treated with lithium. The work of “an unknown psychiatrist, working alone in a small chronic hospital with no research training, primitive techniques and negligible equipment was hardly likely to be compellingly persuasive, especially in the United States,” as Cade put it.[65] And so Cade’s report on lithium, published in an obscure Australian medical weekly, slumbered on.

In 1952, Mogens Schou, a young Danish psychiatrist with a research post at the psychiatric hospital of Aarhus University, was searching for an appropriate biochemical subject on which to do research. The head of the hospital, psychiatry professor Erik Stromgren, drew Schou’s attention to Cade’s article and suggested that they try to verify the article’s claims at Aarhus. Schou was particularly attracted to the theme given that manic-depressive illness ran in the Schou family and that his father, also a psychiatrist, had been most interested in the subject. Schou decided to carry out a placebo-controlled, double-blind trial, one of the first in psychiatry. (In such a trial, a control group receives a sugar pill; neither the patients, the controls, nor the assessing psychiatrists know who has received the lithium and who the placebo.) Schou’s study confirmed Cade’s claims: Lithium provided “symptomatic” relief of mania, meaning that when the treatment stopped, the patient relapsed.[66] Schou later said, “Perhaps more than most

scientists I have been granted the privilege of reaping the fruits of my labor. A number of family members have been given lithium treatment with signal effect [Schou himself was among them]; they might have been hospitalized or dead today if prophylactic treatment had not come around."[67] Schou's study launched lithium into the international world of psychiatry.

Yet the launch was mightily protracted, for only in 1960 were the first North American studies of lithium conducted (one of them in Ewen Cameron's Allan Memorial Institute in Montreal).[68] The Food and Drug Administration approved it in 1970 only after one Oregon psychiatrist threatened to practice civil disobedience and prescribe it anyway, saying that a doctor's duty to treat his patients responsibly took precedence over the government's power to regulate drugs.[69] Not until the 1970s, therefore, did lithium really take off in the United States.

Why a two-decade delay in the adoption of lithium? For one thing lithium was a plentiful natural substance and had no industry backer. Hence no drug company was rooting for it. It also had to confront entrenched therapeutic nihilism on the part of the analysts in the United States and at the Maudsley Hospital in London, the premier training center for British psychiatry. Both Aubrey Lewis, professor of psychiatry at the Maudsley until 1966, and the pioneering epidemiologist Michael Shepherd considered lithium to be "dangerous nonsense."[70] (The British resisted chlorpromazine less because the firm May and Baker was flogging it.) Thus for needless years, patients with mania languished.

Schizophrenia and mania affected at best 2 percent of the population. Depression, by contrast, was a massive illness affecting as many as one individual in four. A drug to treat depression successfully would have almost the same appeal as aspirin. In the history of the cornucopia, it was now depression's turn, and news of a drug called imipramine came soon after Cade's announcement of lithium. Imipramine was the first drug in the history of psychiatry to act specifically against depression.[71]

In 1950, the J. R. Geigy pharmaceutical firm in Basel asked staff physicians at the Münsterlingen asylum in Switzerland to see if an antihistamine that Geigy had developed might serve as a sleeping pill. The pharmacological air was thick with excitement about antihistamines at the time. The Münsingen psychiatrists found that the drug had few hypnotic qualities, yet in their reply to Geigy speculated that it might serve as an antipsychotic, a suggestion the firm ignored.

One of these staffers was Roland Kuhn, then 38, a tall, distinguished, and cultivated psychiatrist who combined an exceptional grasp of the humanities with a background in biochemistry. As part of his medical training in Berne, Kuhn had done an elective in organic chemistry. He had received his psychiatric training at the Waldau asylum under the tutelage of Jakob Klaesi (of the deep-sleep cure). Kuhn had pioneered in Switzerland the use of electroencephalography for studying the brain's electrical activity. Yet at the same time he was, in his early years, an adept of psychoanalysis. He had undergone a training analysis and was a good friend of Ludwig Binswanger at the nearby Bellevue private clinic in Kreuzlingen. Kuhn also attended meetings of Max Müller's "Psychology Club" at the asylum in Münsingen. For Switzerland, it was not a contradiction that someone like Müller could champion the early shock treatments and at the same time be interested in psychology. Switzerland was probably the world epicenter of both psychodynamics and organic psychiatry in the years between 1933 and the 1950s, and Kuhn was right at its heart.

Then Kuhn had one of those road-to-Damascus-type experiences. He had been treating with psychoanalysis a young woman whose complaints seemed to be of a "neurotic-hysterical" nature. He made great progress with psychodynamic therapy, "bringing out unconscious material that corresponded exactly to Freud's theories. Everything went beautifully and she was cured.

"A few days later she came to my office again, garishly made-up and perfumed, with costume jewellery hanging all over her, dressed in loud colors ... and demonstrating irritable and euphoric mood changes, pressured speech and flight of ideas." Kuhn then realized he had made a mistake. The correct diagnosis was mania. He had falsely ascribed a spontaneous recovery to his psychoanalytic "cure." As so often happened in those days, Kuhn had also missed her earlier depression, giving her a misdiagnosis of "hysteria." For Kuhn, manic-depressive illness was an organic condition having little to do with Freud's ideas.

Yet as Kuhn fell away from psychoanalysis, he asked himself, what can we do to help patients like this? Admitting her to the mental hospital for ECT seemed a bit much. "How often I said to myself, 'We should improve the opium cure!' But how?"

Then in 1952, the discovery of chlorpromazine became known. Kuhn and his colleagues put depression and mania at the back of their minds for the

moment. The Münsterlingen staff received free samples of chlorpromazine from Rhône-Poulenc for trials on their schizophrenic patients, but the asylum's limited budget made them unable to order large amounts for routine use. In February 1954, the Münsterlingen staff asked Geigy to send some of the previous antihistamine back for trials on their psychotic patients. The substance (G 22150) turned out to be of some benefit but had unacceptable side effects—and was useless on their depression patients—so they returned it.

At this point, Kuhn was heading up Münsterlingen hospital's pharmacological initiatives. (Only in 1971 did he become director of the hospital.) Kuhn asked Geigy—or Geigy asked Kuhn, according to some accounts—if the hospital might try another drug in the antihistamine series, one with a chemical side chain exactly identical to chlorpromazine's (G 22355 was Geigy's internal code for the drug). The staff tried it on patients with schizophrenia. It made many of them worse, converting quiet chronic patients into agitated whirlwinds of energy. Kuhn consulted with Geigy scientists on what the drug could possibly be doing to procure such bizarre effects, and sometime in 1955 the decision was made to give it to some depressed patients. The response was "absolutely incredible, so exciting," electrifying both the hospital staff and the Geigy scientists who had been following all this with bated breath. Kuhn and the Geigy people had obviously discovered a drug that could relieve depression.[72]

In the first 40 depressed patients who had received it, some of the recoveries were dramatic. Kuhn: "The patients become generally more lively; their low depressive voices sound stronger. The patients appear more communicative, the yammering and crying come to an end. If the depression had manifested itself in a dissatisfied, plaintive, or irritable mood, a friendly, contented, and accessible spirit comes to the fore. Hypochondriacal and neurasthenic complaints recede or disappear entirely." Kuhn told of patients who now were ready to jump out of bed in the morning, to socialize easily with fellow patients, "to amuse themselves and take part in the general life of the hospital, to write letters and interest themselves again in their family circumstances." At visiting hours, the patients' relatives were astonished at the change, declaring, "We haven't seen him this well for a long time." The patients themselves spoke of a "miracle cure."[73]

Depression before the introduction of imipramine (Tofranil) in 1958, the first

drug specifically for the treatment of depression. Here George Cruickshank's 1823 caricature "The Blue Devils" (courtesy of The William L. Helfand Collection, Philadelphia Museum of Art).

The language in which Kuhn reported the transformation is interesting, because it illustrates how resurrectionlike the recovery from depression can be, a recovery that each new generation of antidepressent drugs believes that it alone has achieved; witness the resurrectionist rhetoric accompanying the introduction of the drug Prozac (see pp. 320-324). Kuhn announced the drug at a meeting of the 2nd International Congress of Psychiatry in Zurich in September 1957. There were 12 people in the audience.[74]

In the spring of 1958 Geigy named the compound imipramine (Tofranil). Imipramine was the first of the "tricyclic" antidepressants, so named because

of their three-ring chemical structure (Chlorpromazine has almost the same structure, differing only by two atoms). Given the great utility of antidepressant drugs, rival tricyclics flooded the market, the Merck Company, for example, bringing out amitryptiline (Elavil) in 1961. By 1980 American physicians were writing 10 million prescriptions a year for antidepressants alone, the great majority of them tricyclics. There would be several dozen brands to choose among.[75]

As the cornucopia opened its bounty, psychiatry acquired a new sense of confidence. Psychiatrists were becoming actually able to make people better. "With the introduction of modern treatments, it became relatively easy to remove symptoms," said Felix Post, a veteran of the Maudsley who specialized in geriatic psychiatry and was now thrilled that he could help elderly patients with paranoid ideas or avoid institutionalizing those with chronic depression (who formerly might have suicided if not admitted). As Post put it, "In the past one had to carry psychiatric outpatients over many months or years with bromides and very wearying supportive interviews: 'I.S.Q.' [In Status Quo] was usually the last or only entry in the progress notes. Now most of these patients could be significantly improved...." The availability of these new drugs had therefore transformed the experience of psychiatry of Post's generation: "I started as a lone doctor," he said, "bewildered and frightened by the multitude of apparently hopelessly ill and deteriorating patients. I end as a member of a professional team and with the certainty of being able to help to an important extent almost all my patients."[76]

The discovery of the antipsychotics, antimanics, and antidepressants, drugs that formed the pharmacological basis of the second biological psychiatry, was owing to a mixture of scientific preparation and dumb luck. Yet the scientific interest they generated, and the profits that accrued from their sales, contributed to putting psychiatry on a much more solid scientific footing than it had previously been. This was a footing in the neurosciences.

Neuroscience

The neurosciences endeavor to make major psychiatric disorders understandable in terms of the chemistry and anatomic pathology of the

brain. The conviction that this was feasible underlay the first biological psychiatry, although its representatives made little actual progress. As the young Heinrich Laehr, then an assistant physician at the asylum in Halle, said in 1852, “Insanity is nothing else than a disease, and only medical treatment can prevail against it.” For Laehr, it was the chemistry of the brain that kept people in psychic equilibrium: “An extremely small chemical and physical change in the brain ... will suffice to bring out a mental disorder.”[77] This differs scarcely a whit from the explanation doctors would give patients 150 years later to the effect that, “You are suffering from an imbalance in brain chemistry.” Yet Laehr and other psychiatrists, hunched over their microscopes for hours in the asylum, got nowhere. The task of understanding the brain was too vast for clinicians alone to undertake. In 1891, the Scottish psychiatrist Thomas Clouston adumbrated a grander vision of the neurosciences: “[Some day] it may be possible to take a larger, a more comprehensive, and a more physiological view of the whole subject of brain growth and development in its combined physiological and pathological aspects.” Clouston even envisioned a neuroscience that could “classify the whole of the neuroses of development into one large and most interesting scheme—so giving a physiological coherence to what have been hitherto unrelated pathological facts.”[78] By “neuroses of development” Clouston meant what would later be called schizophrenia and other disorders of brain growth. Yet for decades to come almost no progress was made on the difficult subject of brain chemistry in illness.[79]

The first comprehensive center for investigating the brain, mind, and their disorders was Kraepelin’s Munich institute. Working alongside such researchers as the neuropathologist Walter Spielmeyer and the geneticist Hans Luxenburger, Felix Plaut took the first steps in the discipline of neuroimmunology, documenting the brain’s immune reaction to syphilitic infiltration.[80] With the Nazi rise to power, this great institute lost its cutting edge. Plaut, for example, was driven in 1936 from his post as director of the Serology and Experimental Therapy laboratory to exile in London, where he suicided in 1940.[81]

After World War II, these threads were picked up again in other countries, so that by the time the first International Congress of Psychiatry met in Paris in 1950, there was a critical mass in the sciences underlying biological psychiatry. Each of the neurosciences on which the second biological psychiatry would later repose was present: Denis Hill, then a senior lecturer

at the Institute of Psychiatry of the Maudsley Hospital, talked about the use of electroencephalography in psychiatry; he had pioneered the device in the study of schizophrenia.[82] The functional psychoses, he said, showed "a bewildering variety of anomalies and abnormalities" yet nothing specific for any given illness. (The technique would later be used for identifying developmental delay in child psychiatry.[83]) Max Reiss of Bristol, a refugee scientist from Prague and one of the founders of psychiatric endocrinology, talked about thyroid activity in psychiatric illness as measured with radioactive isotopes.[84] In 1950, Derek Richter had just arrived at Carshalton as the director of the Neuropsychiatric Research Centre and would go on to become a key figure in establishing the discipline of psychopharmacology in Britain. At the Paris conference, he described use of radiolabeled isotopes to follow metabolic activity in the brain.[85] Not all of these approaches would later turn out to be important in clinical psychiatry, but in Paris in 1950 pathways were sketched out that would permit the systematic investigation of the brain as the physical substrate of the disordered mind. The conference represented the true birth of clinical neuroscience as the term later became understood.

If any single individual should be considered the founder of biological psychiatry in the United States, it is probably Stanley Cobb, the Harvard neurologist who in 1934 founded a department of psychiatry at the Massachusetts General Hospital and therewith himself switched to the field of psychiatry. Cobb, who had extensive training in Europe, was interested in "psychobiology supported by the neurosciences." The words are those of neurosurgeon Wilder Penfield, and Penfield wrote about Cobb in the context of Penfield's wider interest in the life of Alan Gregg, the medical director of the Rockefeller Foundation, who from 1933 on funded research in the basic psychiatric sciences, especially neurophysiology and neurology. Gregg steered enough Rockefeller money to Cobb to permit him to do basic biological research at the Mass General for the next 20 years.[86] Out of Cobb's and Tracy Putnam's research came Dilantin, the first drug effective against epilepsy.[87] And out of Cobb's psychiatry department at the Mass General came a core of researchers such as Eli Robins (see p. 300) who would push forward biological thinking in psychiatry at a time when psychoanalysis was the order of the day.

In 1946, Cobb was part of a small group of researchers who met at the Fairmont Hotel in San Francisco. The meeting was organized by two

California neurologists—Johannes ("J. M.") Nielsen and his student George Thompson (the following year they coauthored the first textbook of biological psychiatry in the United States[88]). From this gathering emerged the premier American organization in the biological field, the Society of Biological Psychiatry.[89] The founders saw themselves as continuing the first biological psychiatry. "The concept of the neuronal basis of psychiatry is not at all new," they noted, citing Meynert, Wernicke, and Flechsig. Yet since those days, the discipline had accumulated an "enormous superstructure of observations and interpretations." Now it was time to locate a brain substrate for all this clinical information about illness: "Those who seek to advance the biological concept wish only to build a foundation under that superstructure." "They wish ... to trace the anatomical structures which make these concepts possible."[90] On both sides of the ocean, then, in the early 1950s small groups of researchers were determined to push the neuroscientific basis of biological psychiatry forward.

Perhaps the oldest component of the neurosciences was psychopharmacology, how drugs affect the brain and mind. The term goes back to a nonmedical usage in the late Renaissance, the cleric Urbanus Rhegius's *Psychopharmakon*, published posthumously in 1548.[91] The nineteenth century was dotted with efforts to use drugs as a means of studying the brain. In 1845, Jacques-Joseph Moreau (called "Moreau de Tours"), on staff at the private Parisian clinic of the recently deceased Esquirol, speculated that hashish could serve as a torchlight "in the mysteries of insanity, taking us back to the hidden source of these disorders that are so numerous, so variegated, and so strange." (Moreau himself took hashish for research purposes).[92] Two decades later, the great French physiologist Claude Bernard articulated the notion of using drugs to study the brain: "Poisons represent a means of analyzing the qualities of the nervous system, a kind of physiological scalpel, much more delicate and subtle than ordinary scalpels...."[93] This was the gist of psychopharmacology: a scalpel with which to delineate and treat disorders of the mind and brain. Then in the 1880s, Kraepelin himself conducted the first systematic research on the effect of drugs upon the brain, coining the term "pharmacopsychology," though he saw no therapeutic application for the technique.[94]

Forty years later, things had changed very little. In 1920, the Johns Hopkins' pharmacologist David Macht, describing "the effect of drugs on

psychological functions," said, "The number of contributions to the domain of what we may be permitted to call 'psychopharmacology' is certainly very meagre."[95]

The discovery of LSD in 1943 touched off another round of efforts to use drugs for chiseling out various psychological processes, in this case inducing psychosis experimentally.[96] Yet LSD became a street drug of abuse and this research led to no clinical payoffs.

With the discovery of chlorpromazine and the other early psychoactive drugs, the study of psychopharmacology came into its own. The physiological scalpel had been achieved. The new discipline set out to determine the mechanism of action of the various antipsychotic and antidepressant drugs. The first push in this direction came from Germany, a land effectively ostracized from the international psychiatric community until the mid-1950s. And this initial impetus itself represented a neat bridge between old and new: In 1956, Wolfgang de Boor, a professor at Cologne and graduate of Heidelberg, wrote a textbook of psychopharmacology.[97] This was the first in the world. At Heidelberg, de Boor had been a student of Kurt Schneider, one of the great modern investigators of schizophrenia. But it was at the urging not of Schneider but of Willy Hellpach that de Boor wrote the textbook. Hellpach had been one of Kraepelin's students at Heidelberg. Indeed, Kraepelin's textbook had inspired the young psychologist Hellpach to go into psychiatry. Now in the early 1950s, sick with revulsion at the recent Nazi past, Hellpach wanted to steer German psychiatry once again toward science. These men were academics.

Yet for the most part, the new psychopharmacology was driven by the drug industry and not by the academy. It was de Boor who in 1957 proposed founding the international college of neuropharmacology (CINP are its Latin initials), but Ernst Rothlin, director of the Sandoz Company, was its first president.[98] It was in the area of psychopharmacology that the private sector made a fundamental contribution to neuroscience. The drug industry funded research into such matters as chlorpromazine's action on the brain because only advances in basic science would permit the design of drugs specifically tailored to block whatever biochemical or anatomical pathways were causing psychotic illness. In fact, one could call psychopharmacology the creation of the drug industry, rather than of the academy or clinicians. In the 1960s and after, the discipline became big business in the United States and Britain, the two countries where it most flourished.

Dopamine and serotonin in particular would be the chief players in this psychopharmacological saga, and to find the origin of interest in them we return to 1952. In that year, Betty Twarog, a recent Harvard PhD working in the laboratory of Professor John Welsh, identified serotonin as a neurotransmitter (they worked with a small sample provided by Abbott Laboratories). A year later, she and Irvine Page of the Cleveland Clinic found serotonin in the mammalian brain.[99]

In 1957, pharmacologist Arvid Carlsson and coworkers of the University of Lund in Sweden discovered that dopamine was a neurotransmitter. [100] Carlsson's name recurs again and again in this story, and he is initially associated with establishing the presence of these chemicals in the brain. This was a first cut at applying neuroscience to psychiatry.

Did chlorpromazine and the other antipsychotics that were rapidly being developed work against dopamine? When Carlsson—now at the University of Göteborg—gave them to mice in 1963, dopamine levels in the mouse brain changed. He did not specifically implicate dopamine at that point, but he thought there was a strong possibility that the dopamine system was a site of action for these drugs.[101]

But what about humans with schizophrenia as opposed to well mice? The suspicion arose that dopamine itself made schizophrenia worse, or even caused schizophrenia. This suspicion was strengthened by the finding that amphetamines, which potentiated the action of dopamine, made the symptoms of schizophrenia worse.[102]

Meanwhile, serotonin became implicated in depression. In the mid- 1950s, researchers at the National Institutes of Health—picking up on earlier English work—came to believe that imbalances in serotonin, or "5-HT" as psychopharmacologists still refer to it (5-hydroxytryptamine), were responsible for some psychiatric illnesses. It's worth following this particular powder trail because it ends with Prozac. Yet at the beginning, these researchers were not interested in serotonin because they thought it was involved in depression but rather in psychosis.

The trail was ignited in 1955 at Bernard ("Steve") Brodie's Laboratory of Chemical Pharmacology at the National Heart Institute. Brodie and team discovered that if one gave a compound called reserpine to animals, serotonin vanished from their tissues, including the brain.[103] This finding was the first hard-core link between biochemistry and behavior.[104]

This link began to be strengthened as various researchers discovered that

the new psychiatric drugs changed the chemistry of the brain. When, for example, a group of English scientists gave imipramine to depressed patients in 1960, their blood levels of serotonin dropped way off. This was the beginning of the discovery of the "reuptake mechanism": Antidepressant drugs caused serotonin to be sequestered somewhere in the body[105] (particularly, as it turned out, in the synapses between the neurons). Eight years later, Arvid Carlsson, who had worked in Brodie's lab in 1955, pinned this down in the chemistry of the brain itself rather than merely reasoning from what was happening in blood platelets, as the English had done: The tricyclic antidepressants prevented the neurons from taking back serotonin once they had released it into the synapse. The less the reuptake, the more remaining in the synapse, the more available to work against depression. Carlsson thus fortified a "serotonin hypothesis" of depression.[106]

While this research on brain chemicals, or neurotransmitters, was going on, the whole mechanism of neurotransmission itself, involving receptors for these chemicals at downstream neurons, was being uncovered. This first tapping toward a theory of neurotransmission was Arvid Carlsson's previously mentioned work in 1963: If drugs modified dopamine, maybe dopamine helped carry the nerve impulse about the brain. Then in 1974, Solomon Snyder of Johns Hopkins University noted that antipsychotics such as chlorpromazine attach themselves to the site of the dopamine receptor, preventing dopamine from acting.[107] The key to therapy therefore must lie in modifying the action of neurotransmitters at the receptor site.

Until the 1980s, research in the field tended to be organized along the lines of one-neurotransmitter, one-disease: the catecholamines, meaning such natural brain chemicals as noradrenalin, were assigned to mood disorders. This was called the "amine hypothesis of depression." It later included serotonin. Such neurotransmitters as dopamine were linked to psychosis; and acetylcholine was paired with dementia.[108]

The one-transmitter-one-disease hypothesis seemed so reasonable (and was, moreover, a perfect marketing concept for the drug companies). Yet correlation does not necessarily mean causation. The one-one paradigm tended to collapse in the 1980s. Highly effective antipsychotic drugs such as clozapine were discovered to have little impact on dopamine metabolism but instead affected serotonin. (Although clozapine was discovered in the mid-1970s, only after 1988 did the implications of it for the treatment of schizophrenia begin to sink in.) As more and more neurotransmitters were

identified—over 40 by the mid-1990s—it became apparent that dopamine and serotonin were only two of many transmitters involved in these complex psychiatric disorders and probably did not play a master role.[109] Yet over the years, the one-transmitter-one-disease hypothesis proved highly fruitful in terms of stimulating research on basic brain mechanisms. Somehow these drugs did work, and their very efficacy opened up the domain of the neurosciences in a way that previous hypotheses had not.

Neuropathology—physical lesions in brain tissue—provided yet another leg of support for the neuroscientific interpretation of psychiatric illness. The anatomic study of schizophrenia had once been considered "the graveyard of neuropathologists."[110] Meynert had wrecked his health peering into his microscope, and subsequent generations of psychoanalytically oriented historians made merry of his attempts to identify an anatomic basis for the disease. But Meynert was right. Later neuropathologists, using sophisticated techniques of which he never dreamed, succeeded in discovering the lesions he was seeking. In the 1970s, it was neuropathology that began to discover the answer to one of the great riddles of schizophrenia: Why, if it was a genetic disease, did about half the people who got it come from families with no history of psychiatric illness? The answer turned out to be that nongenetic anomalies in the womb and obstetrical trauma could also damage the developing brain in a way that made it vulnerable to psychotic illness later in life.

This kind of research went back to the neuropathological work that Kraepelin and Spielmeyer had initiated in Munich and that Rosanoff had spark-plugged in the United States. In 1939, for example, Barney Katz, a student of Rosanoff's at the University of Southern California, compared obstetrical data on 100 male schizophrenics and 100 male controls: The patients with schizophrenia had experienced more obstetrical trauma. After World War II, this kind of histopathological research became taboo under the influence of psychodynamic psychiatry, to be relegitimated in the 1970s. At that point, many students of schizophrenia starting discovering problems in the wiring of the fetal brain during development. Joyce Kovelman and Arnold Scheibel of the University of California at Los Angeles, using techniques far more sophisticated than those available to Meynert, began doing autopsies on the brains of patients with schizophrenia who had died at a veterans' hospital in Los Angeles. Looking at a total of 13,680 neurons from 10 patients, they found that in certain brain areas the neurons of the

schizophrenics were much more disorganized than those of controls, evidently a result of the failure of migrating fetal brain cells to link up to their targets. “It seems unlikely that the disorientation of cell-dendrite patterns which we have described can develop at any time except during embryogenesis,” they noted.[111] Another group of scholars determined that individuals with schizophrenia had large numbers of abnormally sized neurons.[112] None of these events could possibly be a consequence of the disease because they occur during fetal development.

Viruses may disrupt fetal brain development. A cohort of Helsinki mothers had been exposed to a severe influenza epidemic in the second trimester of pregnancy (when brain development is most rapid). Twenty-six years later, their children turned out to have exceptional rates of schizophrenia.[113] A much larger survey of 40 years of influenza and schizophrenia among children born in Denmark in the years 1910 to 1950 came up with the same findings.[114]

The response of the brain to these inflammations and disruptions of neuron migration and connectivity is called gliosis: a proliferation of glial cells. As early as 1972, researchers began finding the kind of gliosis that is associated with fetal maldevelopment in adult schizophrenics at autopsy. In 1982, Janice Stevens at St. Elizabeths Hospital in Washington conducted a study of the brains of 25 schizophrenic patients who had died in the hospital, compared with a similar number of brains of non-schizophrenic patients at St. Elizabeths and of nonpsychiatric patients at a general hospital. She found substantial gliosis, particularly in those brain areas where other studies had showed changes in schizophrenia. The gliosis was not a result of ECT or drugs.[115]

Finally, what Meynert did not have at his disposal were the postmodern methods of brain imaging, particularly computed-tomography (CT), magnetic resonance (MR), and positron-emission-tomography (PET) which became available in the 1970s and after, permitting overviews of the brain of extraordinary clarity. In 1976, Eve Johnstone, Timothy Crow and other researchers at the Clinical Research Centre in London-Harrow did a CT study of 17 schizophrenic patients in the hospital, compared with a group of age-matched volunteers who worked at jobs similar to those of the patients before they fell ill. It became apparent at once that the brains of a number of the schizophrenics had a different topography than the non-schizophrenics. Particularly their cerebral ventricles, the reservoir of cerebrospinal fluid in

the brain, were enlarged, the degree of enlargement directly proportional to the degree of impairment.[116] This enlargement was the result of the failure of the tissue around the ventricles to grow.[117] Meynert, laboring at his workbench on microscopic sections, had been unable to gain an overview of such structures as the ventricles. But the imaging techniques of the late twentieth century were able to open up the brain to an extent never before imaginable.

By the mid-1990s, the evidence of neuropathology pointed overwhelmingly to schizophrenia as an organic disease. As neuropharmacologist Floyd Bloom concluded in 1993 (he was at the time at the Scripps Research Institute in La Jolla, California, later editor-in-chief of Science magazine), "The wealth of evidence attesting to a substantial neuropathology in schizophrenia leaves few doubters."[118]

Kraepelin's "dementia praecox" had cast the net too widely. As an organic brain disease, schizophrenia seemed to have two faces. One face, neurodevelopmental in nature, affected males more than females in their teens and early twenties. The other face was probably not neurodevelopmental (the cause remaining uncertain) and struck at mid-life, affecting females more than males.

Yet the accent was on the growing brain. Early-onset schizophrenia seemed to occur in a kind of "two hit" model affecting the brains of fetuses and small children. The first "hit" was genetic, the second uterine trauma or even some later biological or psychosocial life event such as a difficult birth.[119] These events would leave behind physical and behavioral traces, such as smaller heads or the inability to play with other children, that represented fossilized evidence of pathology at the beginning of life.[120] In the mid-1990s, the causes of this failure of development were still unclear. It could be a virus, or genes switching off prematurely. Or, as Bloom put it, "the genome [genes] of a given family tree might 'simply' fail to maintain the expression of one or another of the genes required to complete the process of cortical neuronal migration."[121]

In the 1980s, brain imaging and other neuropathological techniques turned up an organic substrate in a number of psychiatric illnesses apart from schizophrenia. Enlarged ventricles also occurred in manic-depressive illness and were not unique to schizophrenia, suggesting that it too may arise during gestation.[122] From 1987 on, a string of researchers found anatomic and

physiological changes in obsessive-compulsive disorder, an illness that psychoanalyst Otto Fenichel had explained in 1934 as “the ego hav[ing] already begun to adopt protective measures at the time of the original anal sadistic level of libido organization, so that the patient never reached the phallic oedipus complex.”[123] It is hard to imagine more differing explanations.

Which was right? Was it the failure of the personality to evolve from the anal sadistic level or was it brain biology that caused obsessive-compulsive disorder? In 1987, one group of researchers attempted to sort this out with the use of positron emission tomography, in which a special camera followed the uptake in the brain of a radioactive isotope attached to glucose: Active parts of the brain would absorb this radiolabeled glucose. In these studies, patients with obsessive-compulsive disorder were given stimuli (“contaminants”) that were known to set off a patient’s ritualistic handwashing or some other form of behavior. A gamma camera then followed the isotope’s pattern of uptake as the handwashing urge seized hold of the brain. (The patients were permitted to wash their hands between each scanning in order to return their symptoms to baseline level before receiving the next contaminant.)[124] By 1994, it had been established that a loop of electrical activity from the frontal lobe to the basal ganglia then back to the frontal lobe provided an important anatomic substrate of obsessive-compulsive disorder.[125] Such patients then improved (and blood-flow patterns changed) after the administration of a psychoactive drug (Prozac).[126]

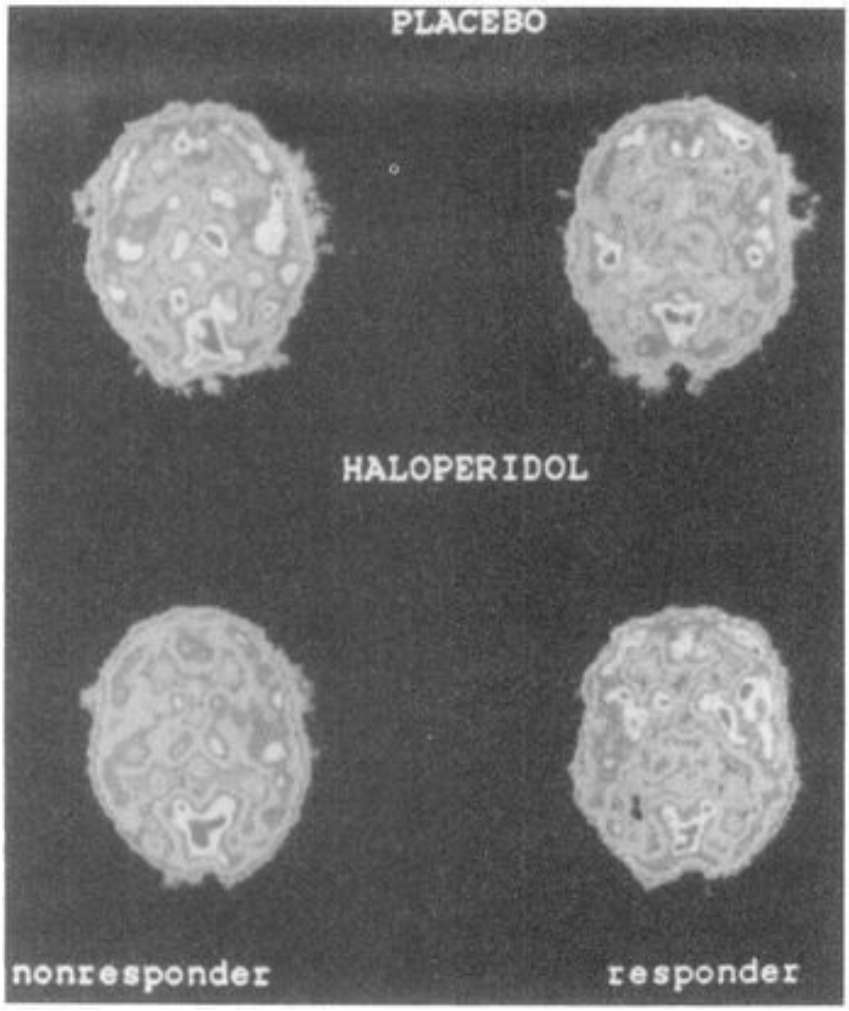
A PET scan (positron emission tomography) of the brains of two patients with schizophrenia on the left and right sides of the page respectively. The upper-right side shows one patient before treatment with the antipsychotic, or neuroleptic, drug haloperidol. (Before being given haloperidol, the patient had been receiving a placebo.) Before the real treatment began, this patient demonstrated quite a low level of metabolism in the deep structures of the brain called the basal ganglia. In the lower right photo, after receiving haloperidol the same patient’s metabolic activity has clearly speeded up: The neurons have lighted up as they absorb a radiolabelled form of glucose.

The left side features a schizophrenic patient who did not respond to haloperidol. The pretreatment image of this patient (upper left) is similar to that of the patient who responded, with pretreatment metabolism being a bit higher. But after this patient is given the drug, he or she fails to respond:

Glucose uptake in the basal ganglia does not increase: The neurons there do not light up.

The significance of this study is that patients who respond well to haloperidol tend to be those with low metabolism in their basal ganglia before treatment: Treatment seems to “nonnalize” metabolic activity.

Source: “Striatal Metabolic Rate and Clinical Response to Neuroleptics in Schizophrenia” by Monte S. Buchsbaum et al., *Archives of General Psychiatry*, 49 (1992), pp. 966-974, photo p. 969. Used by permission. Contrast in the original photo was enhanced because the PET computer prints images in color.



Biological psychiatry thus became able to investigate the causes and

treatments of psychiatric illness by using the scientific method, a method other psychiatrists had virtually abandoned for half a century. Only alliance with geneticists, pharmacologists, radiologists, biochemists, and pathologists made these kinds of advances possible. The Society for Neuroscience, formed of such disciplines, grew from 250 members at its founding in 1971 to more than 11,000 by the late 1980s. Their assignment: to investigate the thousands of gene products made by the brain.[127]

Yet psychiatrists are not really basic scientists; they are clinicians. Part of their mission is making patients feel good while helping them to feel better. They must buttress the human side of the doctor-patient relationship, which is after all an encounter between two human beings, while puzzling about the patient's genetic background, brain anatomy, and the most effective choice of drugs.

In the world of the clinic, it turned out to be unwise for the psychiatrist to take on the airs of a laboratory scientist, and those who did were rebuked in the massive outpouring of rage in the 1960s and after called the antipsychiatry movement. Antipsychiatry went hand in hand with the greatest debacle of twentieth-century psychiatry: the return of patients to "the community," sometimes known as deinstitutionalization. Both may only be understood as unwitting consequences of biological psychiatry.

Antipsychiatry

The history of medicine is full of ironies. One irony is that, in the 1960s and 1970s, the crown of victory was snatched from the physician's head at the very moment that he or she became able to cure organic disease in the body. In these years, primary care in general witnessed a new alienation in the doctor-patient relationship, a consequence of doctors, who knew that they could now cure patients with penicillin, neglecting the psychological side of their relationship with their patients.[128]

Similarly in psychiatry, the advent of effective new medications for psychosis and neurosis may have induced a certain insouciance toward the patients' need to feel cared for. A five-minute interview to check for drug side effects is not the same as the lengthy and persuasive expression of concern that psychotherapeutically oriented psychiatrists tended to display. It

was the very aura of science, which the antipsychiatry movement would always put in mocking quotation marks, rather than custodialism as such that drew the protesters' rage: In the iconography of the movement the psychiatrist would be depicted with antennas on his head contemplating a naked female figure atop an oscilloscope, or wearing a space helmet and grimly preferring an outstretched hypodermic syringe. [129] Somehow, psychiatry's very real shift to science was associated with an imputed loss of caring.

Many other factors as well helped germinate the antipsychiatry movement, not merely changes within psychiatry itself. The whole social climate of the 1960s fostered hostility toward authority, medical and otherwise. Leftist writers saw in psychiatry the controlling arm of "the bourgeoisie," Karl Marx's term for the stock- and bond-holding upper middle classes. Early feminist writers saw in the male psychiatrist the comprador representative of the male gender, extending patriarchal power over women.

The antipsychiatry movement's image of biological psychiatrists.

Source: Madness Network News Reader, 1974, p. 91.

Yet a handful of intellectuals in particular became identified with the antipsychiatry movement.[130] And the force of their ideas brewed up a mass hostility to the advance of biological thinking within psychiatry. The movement's basic argument was that psychiatric illness is not medical in nature but social, political, and legal: Society defines what schizophrenia or depression is, and not nature. If psychiatric illness is thus socially constructed, it must be deconstructed in the interest of freeing deviants, free

spirits, and exceptional creative people from the stigma of being "pathological."[131] In other words, there really was no such thing as psychiatric illness. It was a myth.

Although antipsychiatry movements had flourished throughout the nineteenth century, their late-twentieth-century rebirth began with the virtually simultaneous publication in the early 1960s of a series of exceptionally influential books on psychiatry. Most famous perhaps of these was Michel Foucault's *Madness and Civilization*, published in 1961 (see p. 276), which argued that the notion of mental illness was a social and cultural invention of the eighteenth century. Yet there were several other blockbusters, and collectively they became the intellectual springboard from which the theorists of deinstitutionalization of the late 1960s would launch themselves.

Earliest of the founding fathers—they were all men—was Thomas Szasz, a Budapest-born psychoanalyst, who had trained just after World War II in Chicago. When called to active service in the Navy in 1954, Szasz, who was then 34, used the time to put down on paper a notion that had long troubled him, that mental illness was in fact a "myth," a medical misapprehension foisted on individuals who had problems in living. In his 1960 book *The Myth of Mental Illness*, he called the whole notion of psychiatric illness "scientifically worthless and socially harmful."[132] The book enjoyed wide currency and the American intellectual class began asking, if there is no such thing as mental illness, how can we justify locking people up in asylums?

Sociologist Erving Goffman's work *Asylums*, published in 1961, had an even greater impact on intellectuals. In the context of a fellowship in 1955-1956 at the National Institute of Mental Health, he did field work at St. Elizabeths, an institution that, at the time, had over 6,000 patients. He didn't like what he saw, interpreting the asylum as a "total institution," or closed system that infantilized patients and restricted their lives. "Every social arrangement in a mental hospital seems to point to the profound difference between a staff doctor and a mental patient." On admission, said Goffman, "the recruit [the patient] begins a series of abasements, degradations, humiliations, and profanations of self." As with inmates in prisons, among psychiatric patients "there is a strong feeling that time spent in the establishment is time wasted or destroyed or taken from one's life; it is time that must be written off; it is something that must be 'done' or 'marked' or 'put in' or 'pulled.' ... As a result, the inmate tends to feel that for the

duration of his required stay—his sentence—he has been totally exiled from living." Although many of Goffman's criticisms were justified, again the underlying assumption was that there was no such thing as mental illness and that the pretension of professionals to treat it was nothing more than a shameless power-grab. As for psychiatric illnesses that might justify confining a patient, there were, thought Goffman, none.[133]

The works of Foucault, Szasz, and Goffman were influential among university elites, cultivating a rage against mental hospitals and the whole psychiatric enterprise. Yet the book that did most to inflame the public imagination against psychiatry was a novel written by Ken Kesey. Kesey had just finished taking a creative writing course at Stanford when he volunteered for government LSD experiments conducted at a veterans administration hospital at Menlo Park. He stayed on to take a job as an orderly at the hospital. Out of this experience came his 1962 novel *One Flew Over the Cuckoo's Nest*, a book that formed the image of psychiatry for an entire generation of university students. Kesey's notion of psychiatric illness was embodied in the novel's antihero, Randle McMurphy:

> [McMurphy] says he was just a wanderer and logging bum before the Army took him and taught him what his natural bent was ... they taught him to play poker. Since then he's settled down and devoted himself to gambling on all levels. Just play poker and stay single and live where and how he wants to, if people would let him, he says, "but you know how society persecutes a dedicated man."

Indeed: Society persecuted McMurphy by locking him in jail a few times and then when he proved too troublesome, putting him in a psychiatric hospital.[134] The message: Psychiatric patients are not ill, they're merely deviant. In 1975, director Milos Forman made the novel into a movie that became United Artists' biggest-ever hit at the time. It swept the Academy Awards of that year by winning all five main Oscars.

In 1966, sociologist Thomas Scheff, at the University of California at Santa Barbara, determined that the real problem in so-called psychiatric illness was "labeling." "Most chronic mental illness is at least in part a social role," said Scheff. "The societal reaction is usually the most important determinant of entry into that role." Mental illness itself, which Scheff always put in ironical quotation marks, was nothing more than deviance. When society labels a rule-breaker as mentally ill, the effect is to stabilize the

deviance, to let the "agents of social control" identify him or her as a target. A diagnosis of mental illness, therefore, said nothing about the person, it meant rather that the system was incapable of accommodating deviance.[135] (Of this entire approach, St. Louis psychiatrist Samuel Guze said, "Nearly all of us who spend our lives working with psychiatric patients and their families consider the 'labeling theory' fundamentally ludicrous."[136])

Among the earliest of the antipsychiatry writers in Britain was Ronald D. Laing, author in 1960 of *The Divided Self*. A Scot, Laing had trained in psychiatry at Glasgow and in psychoanalysis at the Tavistock Clinic in London. Like Szasz, he then turned on analysis. In his early work, he proposed a psychology of schizophrenia in which apparently gibberish speech was just designed "to throw dangerous people off the scent.... The schizophrenic is often making a fool of himself and the doctor. He is playing at being mad...."[137] Laing later charged that sick families were the cause of schizophrenia and that the so-called illness in fact represented a gifted and creative state of consciousness, a sane response to a mad society. As he said in 1964, schizophrenics were off "explor [ing] the inner space and time of consciousness." "Perhaps we will learn to accord to so-called schizophrenics who have come back to us, perhaps after years, no less respect than the often no less lost explorers of the Renaissance." "Future generations," said Laing, "will see that what we call 'schizophrenia' was one of the forms in which, often through quite ordinary people, the light began to break through the cracks in our all-too-closed minds."[138] Incredibly, Laing became the chief investigator in the schizophrenia research unit at the Tavistock.

Foucault himself acquired official entree to progressive Anglo-Saxon circles with the publication in 1967 in the *Neau Statesman* of Laing's full-page review of the abridged English translation of Foucault's *Histoire de la Folie*, entitled *Madness and Civilization*.[139] At this point, Foucault became the chief authority on the wickedness of the psychiatric enterprise.

By the end of the 1960s, the antipsychiatric interpretation of "so-called psychiatric illness" had gained the catbird seat among intellectuals both in the United States and Europe. In these circles, a consensus had formed that the discipline of psychiatry was an illegitimate form of social control and that psychiatrists' power to lock people up must be abolished with the abolition of institutionalized psychiatric care, Pinel's therapeutic asylum.

Even though these interpretations were very popular among college students and intellectuals, actual patients found them less convincing. Joanne

Greenberg, author as “Hannah Green” of I *Never Promised You a Rose Garden*, had a real psychiatric illness. She hated the Kesey book. She later said, “Creativity and mental illness are *opposites*, not complements. It’s a confusion of mental illness with creativity.... Craziness is the opposite [of imagination]: it is a fort that’s a prison.”

“Some of the people at the [Chestnut] Lodge were people who were too soft on mental illness in that they thought it was creative and lovely or at least until they had seen enough of it. People would tell you what perceptive things a patient had said. The thing is I want to choose my perceptions. I don’t want them to come out of some kind of unconscious soup. I want it to be something I choose to say, not something that says me.”[140] Yet in the 1960s and 1970s, something that says me was still considered a positive experience, and persons undergoing it were to be sheltered from psychiatry. Above all, they were to be sheltered from the asylum.

Return to “the Community”

Long before the rise of the antipsychiatry movement, the destruction of the asylum had begun. Patients were to be returned to “the community.” That the very phrase now turns to ashes in one’s mouth is evidence of one of the greatest social debacles of our time.

Yet the pre-1960s asylum was not exactly a triumph of civilization. As early as World War II, the public began to learn from the negative reports of conscientious objectors doing service there that mental hospitals had become pretty desperate places. In 1946, crusading journalist Albert Deutsch, who had known Clifford Beers and had been active in the mental-hygiene movement, set out to investigate the state of the country’s asylums. Taking a photographer with him, he toured mainly those “located in or near great centers of American wealth and culture.” Deutsch’s 1948 book *The Shame of the States* reflected the horrors of what he had seen. Gone was the sheltering environment that William White had evoked several decades previously, with its kindly but paternalistic superintendent and family-style staff.[141] Deutsch painted a picture of men and women who had been flung into misery and then seemingly forgotten. “One of the most poignant scenes in a mental hospital—one witnessed on every visit—is that of patients timidly pulling at

a doctor's arm or coat as he rushed through the ward on rounds:

> "'Doctor, can I see you for just a minute?'
>
> "'Sorry, next time, next time,' is the invariable answer.
>
> "'What else can I do?' one doctor recently told me, despairingly. 'I know I should see many more patients individually. But how can I when I have five hundred patients under my care?"'

At the Philadelphia State Hospital for Mental Diseases—known to locals as "Byberry"—Deutsch saw "hundreds of patients sleeping in damp, bug-ridden basements. Noisy and violent patients made life intolerable in barnlike dayrooms because there weren't seclusion rooms where they might be isolated until calmed down." The hospital did not have enough personnel to administer insulin coma therapy. And the photographs Deutsch published of the male "incontinent ward" were truly hair-raising, like "a scene out of Dante's Inferno," as he put it. "Three hundred nude men stood, squatted and sprawled in this bare room, amid shrieks, groans, and unearthly laughter."[142]

The nation was appalled. Early in 1949, 20th Century-Fox released a movie of Mary Jane Ward's partly autobiographical novel about mental illness *The Snake Pit*, starring Olivia de Havilland. *Time* magazine previewed it in December 1948 with a smiling de Havilland on the cover, the face of madness just behind her. "When the superintendent of nurses at a California institution visited the set," *Time* revealed, "she looked at the sobbing, muttering, staring women and said, 'Why, they all look like my own girls.'"[143]

Midst this horrendous publicity for psychiatry, on which the antipsychiatric movement would later feed, several basic realities were obscured. One is that most patients younger than 65 were discharged relatively rapidly from mental hospitals: They did not experience prolonged stays to say nothing of lifelong incarceration. In the years 1946 to 1950 at Warren State Hospital in Warren, Pennsylvania, almost 80 percent of all patients under 65 were released within five years.[144] Second, much of the bizarre posturing and disordered movement that Deutsch and later antipsychiatric writers ascribed to "hospitalism," meaning the iatrogenic results of institutionalization, turned out to be an inherent biological feature of such illnesses as schizophrenia that, in affecting the entire brain, affect the entire nervous system as well.[145] Third, even though conditions in mental hospitals were unsettling enough, there were worse alternatives. One was

being tossed to the mercy of the streets.

Time cover of Olivia de Havilland, starring in “The Snake Pit” released in 1949 (Copyright 1948 Time Inc., reprinted by permission).

What initiated the massive discharge of psychiatric patients to the “community,” a process known as deinstitutionalization, was the introduction of antipsychotic drugs in 1954, the year the Food and Drug Administration licensed chlorpromazine.[146] Once it became possible to calm agitated patients and abolish psychosis with drugs, patients could in theory live quite easily in normal community settings until the psychosis finally just burned itself out.

Henry Brill, assistant commissioner of mental hygiene in New York State, was among the earliest to initiate use of the new drugs. In January 1955, undertaking considerable risk to his own career from hostile Freudians, he began the use of chlorpromazine and reserpine in the New York public hospitals. Within the next 12 months, the decline in the state's resident psychiatric population began.[147] Similar declines were soon precipitated elsewhere. In a strict sense, therefore, deinstitutionalization was a consequence of the second biological psychiatry and not the antipsychiatry movement.

In the United States, the number of patients in state and county mental hospitals declined from its historic high of 559,000 in 1955 to 338,000 in 1970, further to 107,000 in 1988, representing a decrease over the 30 year period of more than 80 percent.[148] The red bricks lost four-fifths of their patients. In 1955, 77 percent of all psychiatric "patient care episodes" occurred in mental hospitals, in 1990 only 26 percent. Amplifying the shift was a fivefold expansion in the total volume of care in mental-health organizations over that period, from 1.7 million episodes in 1955 to 8.6 million in 1990.[149] This was a shift in the locus of care virtually without precedent in the history of medicine.

Yet if drug therapy kicked deinstitutionalization off, what kept it going, driving patients of all kinds into the community whether they were treatable with drugs or not? It was the combined pressure of the antipsychiatry movement outside of medicine and of the ideology of community psychiatry within medicine. The antipsychiatry movement preached that mental hospitals as such were wicked, given that there was no such thing as mental illness. And well-meaning psychiatrists who had absorbed the teachings of Joshua Bierer, Thomas Main and others, believed that "therapeutic communities" could be constituted out there in the cold streets of the big cities, a romanticized vision of welcoming friends and neighbors clasping the mentally ill to their bosoms. The National Institute for Mental Health, created by the Mental Health Act in 1946, was partly responsible for the propagation of this myth. The institute was supposed to administer the Community Mental Health Centers spawned by John F. Kennedy's 1963 legislation. But the CMHC's soon became diverted to psychotherapy sessions for the walking well, and in the first decades of deinstitutionalization no administrative arrangements were made to receive the actively ill patients who were simply being pushed out of the mental-hospital doors.[150]

Deinstitutionalization therefore became the true “shame of the states” in the United States. A third of the homeless were in fact mentally ill, unable to organize their lives and find shelter or work. Other discharged patients drifted into the criminal justice system, one study finding that 14 percent of county-jail inmates had had previous psychiatric treat-meant. [151] The community structures that were supposed to receive them turned out to be nursing homes and boarding houses.[152] And the antipsychotic medications that in hospital had provided such effective relief were often not taken once the patients were on the street because of tardive dyskinesia, the troublesome side effect that caused facial twitches and other involuntary movements.[153] “The street people, among the most helpless of adult human beings, are the natural prey of anyone looking for some loose change, a pack of cigarettes, a bottle,” said one account. “They are rabbits forced to live in company with dogs.”[154]

It was the politics of “discharge and be damned,” said a British psychiatrist returning to the United States decades after having worked there in the early 1950s. “At least when I visited them [during my first trip] they had a home, inadequate as it was in many ways, and they were cared for, although the standard of care may not have been of the best. But despite all the shortcomings, life for those hapless, helpless, chronically sick people must be infinitely worse now that there is no alternative to the sidewalk, the doss-house, or the prison.”[155]

In the 1980s, a backlash to these desperate conditions took place, as general hospitals and private psychiatric hospitals began admitting increasing numbers of psychiatric patients. The total number of private psychiatric hospitals in the United States increased from 150 in 1970 to 444 in 1988.[156] Of the 1.6 million Americans admitted to psychiatric institutions in 1994, 43 percent were in a general hospital, 35 percent in a state or county asylum, and 11 percent in a private mental hospital.[157] Ultimately therefore, the antipsychiatry movement failed. And community psychiatry though worthy in spirit became discredited as a practical means of treating serious psychiatric illnesses that, not having arisen in the community, could not be cured in it either.

The Battle over ECT

Before antipsychiatry faded, however, it left a last poisoned dart lying in the path: hostility to electroconvulsive therapy, or ECT. Throughout the 1950s, ECT had been regularly practiced. It counted merely as one psychiatric treatment among many and had no particular public profile. The *Reader's Guide to Periodical Literature*, for example, listed only one article on the subject in the entire decade, in *Science Digest* by journalist Lucy Freeman asserting that, "despite some amazing results ... we're overdoing shock treatments."[158]

All this changed dramatically with the advent of the antipsychiatry movement in the early 1960s. The antipsychiatrists charged that ECT damaged the brain, that it was used as a form of discipline rather than therapy, and that it was therapeutically useless in any event. (It was in fact true that state asylums sometimes threatened patients with ECT to keep them in line, witness the "Georgia Power Cocktail" at Milledgeville.[159]) Yet since there was little evidence for the assertions about brain damage and therapeutic nullity, the ferocity of the opposition to ECT—for hostility to it became a kind of religious crusade—must have been based on something else. In retrospect, it was probably a cultural squeamishness about passing electricity into the body based on association with electrocution and the death penalty. There is no mistaking the hostility that ECT generated among the antipsychiatrists. Goffman, for example, referred to it scantingly in *Asylums*, where the sinister psychiatrists refused to show patients the ECT room in advance.[160]

Yet it was Kesey's book above all that identified ECT with psychiatric wickedness in the public mind, referring to "that filthy brain-murdering room that the black boys [the orderlies] call the 'Shock Shop.'" Here is what happened to poor "Ruckly" after they gave him ECT in the Cuckoo's Nest:

> They brought him back to the ward two weeks later, bald and the front of his face an oily purple bruise and two little button-sized plugs stitched one above each eye. You can see by his eyes how they burned him out over there; his eyes are all smoked up and gray and deserted inside like blown fuses. All day now he won't do a thing but hold an old photograph up in front of that burned-out face, turning it over and over in his cold fingers, and the picture wore gray as his eyes on both sides with all his handling till you can't tell any more what it used to be.[161]

That this account had nothing to do with what actually happens in ECT was

beside the point. It scared the wits out of a generation of readers: ECT must be stopped.

There was one more factor, too. ECT became an object of enmity in the eyes of the Scientology movement. L. Ron Hubbard, who launched "dianetics" in 1950 as an alternative to psychotherapy (Scientology was established as a "church" in 1954), opposed ECT from the very beginning. ECT, he felt, put new and poisonous "engrams" into the body, which it was the task of dianetics to remove if the brain had not been too badly damaged.[162] As the Scientology movement became wealthy and powerful, its Citizens Commission on Human Rights spearheaded campaigns to make ECT illegal. Thomas Szasz was the organization's first consulting psychiatrist.[163]

Randle P. McMurphy (Jack Nicholson) and Chief Bromden (Will Sampson) await a session in the "shock shop" in a scene from the 1975 film adaptation of *One Flew Over the Cuckoo's Nest*. The hit film's portrayal of ECT as a punishment for deviant behavior both reflected and contributed to the anti-ECT sentiments of the period.

Source: One Flew Over the Cuckoo's Nest. Directed by Milos Forman, produced by Saul Zaentz and Michael Douglas. Copyright © 1975 Fantasy Films. All rights reserved.

Responding to agitation by patients' rights groups, American state legislatures began regulating ECT. Utah in 1967 was first to pass such legislation, and by 1983, 26 states had passed some kind of statute, six others issued administrative regulations, and one state was operating under federal court orders.[164] In some jurisdictions, ECT was actually banned. In the fall of 1974, the California legislature passed a bill that was tantamount to prohibiting ECT, stipulating that, even when a patient volunteered for it, it could only be administered with the consent of a review panel appointed by community medical authorities, and only after all other psychiatric remedies had been exhausted. Violation could lead to revocation of a physician's license. Challenged in the courts, the legislation did not take effect.[165] In 1982, a Coalition to Stop Electroshock in the city of Berkeley gathered 1,400 signatures necessary to put a measure banning ECT on the ballot ("a misdemeanor punishable by a $500 fine or six months in jail"). It passed by a wide margin but again, the ban was overturned. [166] With the backing of the Scientologists, in 1995 legislation to ban ECT was introduced in the Texas

legislature.[167] The bill did not pass but received wide press coverage. Given this kind of public alarm, it is little wonder that ECT virtually vanished from university training programs in the years from 1960 to 1980. Psychiatrists trained in this era tended to be unfamiliar with it. In Monroe County, New York, for example, the average annual incidence of first-time ECT dropped from 31 per 100,000 population in 1961 to 19 in 1975, a decline of 39 percent.[168]

The psychiatrists' counterattack began in Massachusetts in the early 1970s when the antipsychiatrists attempted to get the state legislature to ban ECT.[169] Early in 1972, Milton Greenblatt, the State Commissioner of Mental Health, convened a task force on "electroconvulsive therapy in Massachusetts," chaired by Fred Frankel of Harvard University. Frankel, whose research interests were in hypnosis and psychosomatic medicine, fell between the state's two warring camps, the psychoanalytically oriented psychiatrists in private practice and the institutional psychiatrists. The task force polled 650 psychiatrists in the state, of whom 66 responded, the balance of opinion being mildly favorable to the procedure. In their 1973 report, task force members betrayed a decided fondness for psychotherapy in any condition short of suicide. Yet they did manage to cough up a reserved approval for ECT in major depression and mania, though the hedges were high: How depressed did a patient have to be before qualifying for ECT? "The use of the term 'depressed' by the patient should not immediately be construed to imply the presence of an affective disorder." The principal advantage of shock therapy, they reasoned, lay in "help[ing] a patient re-establish lost control so that he can enter into a therapeutic relationship."[170]

In imitation of the Massachusetts task force, in the fall of 1974 the American Psychiatric Association decided to convoke its own task force on ECT and made Frankel the chair. The task force polled the APA membership, about a third of whose members were unfavorable to its use. Yet the Association's members who did use ECT reported the outcomes to be quite good. Moreover, 83 percent of the patients on whom these psychiatrists reported declared themselves satisfied with the results of treatment. As for the side effects, of which so much had been made, fewer than half the patients had experienced even temporary loss of memory of events surrounding the ECT, 13 percent "temporary loss of distant memories," and 1 percent "permanent loss of distant memories." How did ECT compare with drug treatment? In its 1978 report, the task force found the two treatments evenly

balanced, preferring ECT for severely depressed and suicidal patients (given that antidepressants took about three weeks to act). Though hedged in the language of dubiety and caution, overall the report gave a green light to ECT in the major psychiatric disorders, particularly depression.[171]

The hedges began to come away in the early 1980s when Max Fink emerged as an energetic advocate of ECT. Fink, one of the early leaders in psychopharmacology while at Hillside Hospital in the 1950s, had later become professor of psychiatry at the Stony Brook campus of the State University of New York. He first encountered ECT as a resident at Bellevue in 1944 and had employed it ever since. In 1981, Fink argued that all available data made a "compelling" case for the superiority of ECT over antidepressant drugs. The main reason for not having recourse to ECT, he said, might be "the public image of its special hazards."[172] In other words, two decades of meddlesomeness by ECT's primarily lay opponents had resulted in scaring the public away from the treatment of choice in a major psychiatric disorder.

A full-scale effort to rehabilitate ECT now began. In the summer of 1985, the National Institutes of Health convoked a "consensus development conference," bringing together experts from all over the neurosciences to determine the benefits and dangers of ECT compared with other treatments. "Not a single controlled study has shown another form of treatment to be superior to ECT in the short-term management of severe depressions," the report concluded. And the risks? "Overall, the risk is not different from that associated with the use of short-acting barbiturate anesthetics." The commonest side effect was loss of memory for a few weeks surrounding the treatment. The report concluded that medical schools should restore training in ECT to the graduate curriculum in psychiatry.[173]

In the 1980s, psychiatric organizations in many countries began coming out in favor of ECT.[174] The begrudging document the American Psychiatric Association had issued in 1978 now sounded out of date, and in 1987 the APA constituted a new task force on electroconvulsive therapy. Their report appeared in 1990. No longer was ECT just for desperate cases on the verge of suicide. "ECT is an effective treatment for all subtypes of unipolar major depression," for manic-depressive illness (now called bipolar disorder), mania, and psychotic schizophrenia, the report concluded.[175] By 1994, psychiatry residents would be able to carry with them in the pockets of their white coats a compact guide listing the amount by which ECT improved on

other treatments in depression: ECT versus simulated ECT, a 31 percent improvement; ECT versus placebo, 41 percent; ECT versus mainline antidepressants ("heterocyclic antidepressants"), 20 percent, and so on.[176]

The rehabilitation of ECT did not mean that it would be universally used in psychiatry but merely that it had ceased to be an object of horror. By 1988, according to one survey, fewer than one psychiatrist in ten had used it in the previous month.[177]

In the patients' culture, a pool of gentler anecdotes had started to fill, in contrast to scare stories about the cuckoo's nest. When psychologist Norman Endler was himself stricken by depression, he and his doctor tried everything in the book, to little effect. In the depths of the illness, Endler's behavior became "bizarre and paranoid," as he later put it after his recovery. There was the episode of the urine bottle: The doctor had requested a urine sample and Endler's wife, after searching the house, could find only an empty medicine bottle that had contained their daughter's antibiotic. "I was reluctant to use it because for some reason it was impossible to remove the label completely. It was possible, however, to remove all identifying information. This was not good enough for me. For some reason I was convinced, irrationally, that if I left a sample in that container Dr. P. would think that it was not my urine sample and would put me in the hospital for lying." Endler ran about the house refusing to leave the urine sample. "This upset everyone in the house and I remember my wife yelling at me."

Finally Endler's doctor recommended ECT. Endler, a psychologist, was at first horrified. He and his fellow psychologists had drunk deeply of the antipsychiatry lore of the 1960s. In the past, he had argued with psychiatric colleagues, showing them negative articles and accusing them "of getting a 'charge' out of it." "Most of my psychological colleagues had negative attitudes that were no different from mine. We were all biased and prejudiced and refused to accept the new techniques."

But now Endler was desperate. He said he would have walked naked down the city's main street if his psychiatrist told him that would help. That evening a fellow psychologist came over to remonstrate with him. "He wanted to know why I was willing to 'get half my brains burned out' and why I wasn't getting deep psychotherapy."

On Tuesday, August 30, 1977, Endler agreed to ECT as a course of treatment. He booked into the clinic that Friday morning, lay down on a cot and was wheeled into the ECT room. "A needle was injected into my arm and

I was told to count back from 100. I got about as far as 91." The next thing Endler knew, he was awake and in the recovery room. "I was slightly groggy and tired but not confused. My memory was not impaired. I certainly knew where I was." He rested a bit, was given some cookies and coffee, and went home. Over the two weeks ahead, Endler received six more treatments. He felt better with each successive one. By September 15, he was able to resume the chairmanship of his department full-time. "A miracle had happened in two weeks," he said. "I had gone from feeling like an emotional cripple to feeling well."[178]

These are not trivial achievements. The ability of the new biological psychiatry to make individuals like Norman Endler better represents an accomplishment of historic dimensions. Yet it opened a schism within psychiatry between the "tough-headed" types, characterized by a fondness for medical models, organic explanations of illness, and pharmacotherapy, and "soft-headed" types, who invoked "psychosocial" models of illness, explained symptoms as the result of problems in living, and inclined to psychotherapy and family therapy. Whereas the soft-headed practitioners were represented by psychoanalytic, family, and existential schools of thought, the tough-headed identified with biological psychiatry. [179] "There is no such thing as a psychiatry that is too biological," said Samuel Guze, one of the toughest of the tough-headed.[180]

Yet is was not part of the biological brief to argue that all psychiatric illnesses are attributable to discreet, identifiable brain lesions. Social and psychological factors clearly play a role in the genesis of dysphoria, not merely by triggering an underlying genetic predisposition but by independently converting stress and despair into illness. As a leading biological psychiatrist said, "It would be clinical folly to underestimate the importance of psychological and social factors in the manifestations of mental illnesses or to overlook the psychological aspects of the ... biological therapies."[181]

Yet in the 1950s, advocates of psychoanalysis and community psychiatry argued that biology played virtually no role, that it was all nurture and no nature. So the argument that nature and nurture stand in some kind of fifty-fifty relationship is already quite extreme from the viewpoint of 30 years ago. One need not be an organic absolutist to place organic factors in their proper perspective: Nature and nurture intertwine. But even winning for nature a place at the table has proven a capital achievement.